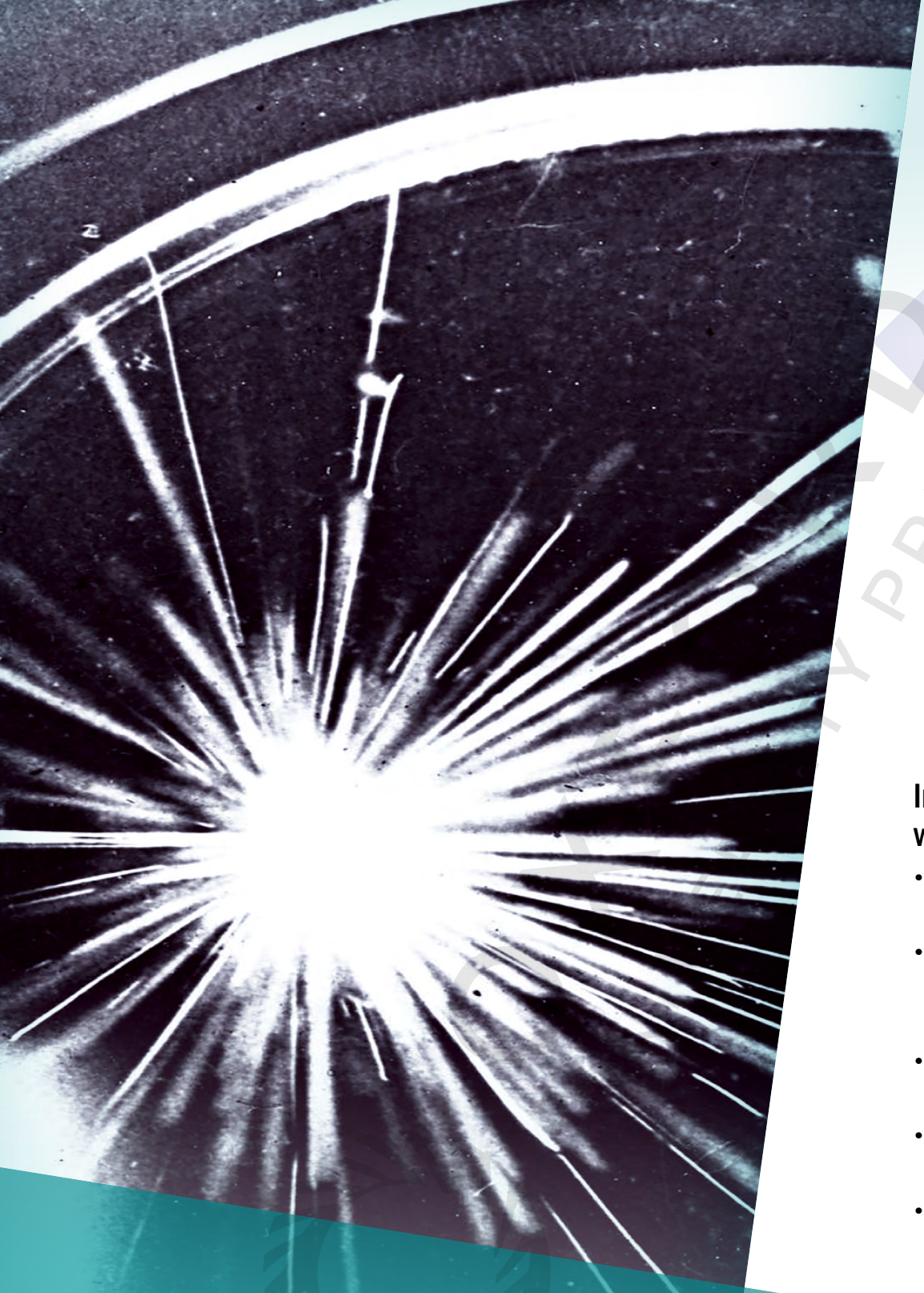


1

Radiation and Radioactivity

In this chapter, we will learn about

- the production, properties and applications of X-rays
- the safety precautions in handling radioactive materials
- the detection of nuclear radiation
- the nature and properties of nuclear radiation
- radiation safety

1.1 Ionizing radiation and X-rays

The term 'radiation' is usually associated with dangers and disasters when it comes up in the media.

Flipped classroom
Watch this and answer the questions.

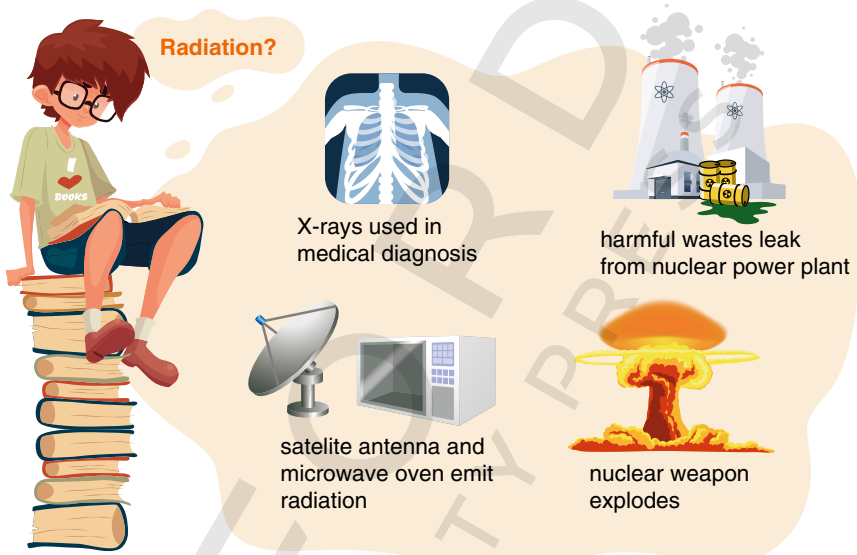


Fig 1.1a *What comes up in your mind when thinking of radiation?*

Cross-link
Radiation is also a process of heat transfer, which is discussed in Book 1 Ch 2.

▶ However, in physics, radiation is just the energy emitted by an object in the form of waves or particles. It may or may not be harmful.

1 Ionizing radiation

Radiation can be categorized into **ionizing radiation** and **non-ionizing radiation**.

1 Ionizing radiation is energetic enough to remove electrons from molecules (Fig 1.1b). The electrons being knocked out may be captured by neutral molecules nearby. The molecules that lose or gain electrons become charged **ions**, i.e. become **ionized**.

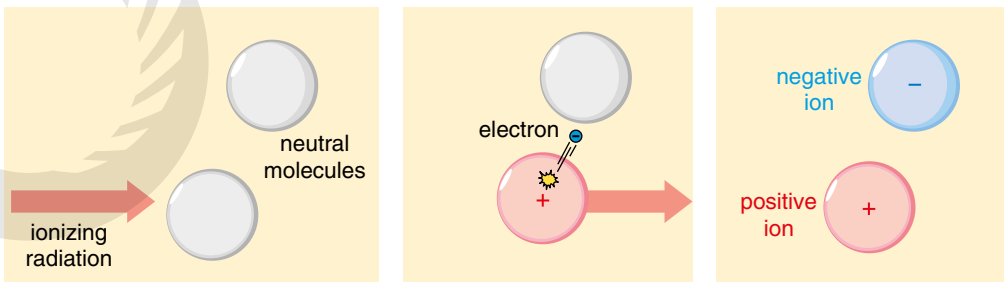


Fig 1.1b *Ionizing radiation produces an ion pair.*

2 Non-ionizing radiation cannot ionize molecules.

In this book, we will focus on ionizing radiation.

ionizing radiation 致電離輻射　non-ionizing radiation 非電離輻射　ion 離子　ionize 電離

Cross-link
You learned EM waves (radiation) in Book 3B Ch 6.

High-frequency electromagnetic (EM) radiation, including **X-rays**, is ionizing while low-frequency EM radiation is not (Fig 1.1c).

Some high-energy UV radiations are also ionizing.

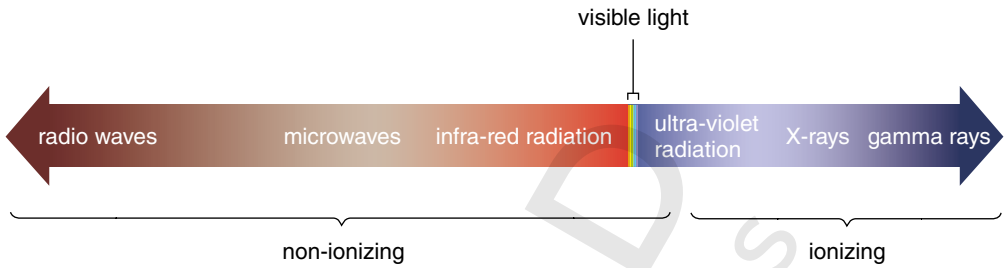


Fig 1.1c *Ionizing and non-ionizing electromagnetic radiation.*

Ionizing radiation can damage living cells and is highly dangerous. Its effects on our health will be discussed in Chapter 1.3.

DSE goal
Realize X-rays as ionizing EM radiations of short wavelengths with high penetrating power.
Realize the emission of X-rays when fast electrons hit a heavy metal target.
Discuss the uses of X-rays.

DSE exam
12(1A)Q34, 22(1A)Q31

2 X-rays

In 1895, *Wilhelm Röntgen* discovered a new kind of invisible radiation. He called it X-ray, where 'X' stands for 'unknown'. Nowadays, X-rays are not mysterious anymore and are widely used for different purposes.

a Production

Today, we use an X-ray tube to produce X-rays (Fig 1.1d). In the tube, electrons are emitted by a hot filament (– terminal) and are accelerated by a high voltage towards a **heavy metal** target (+ terminal). When the **fast electrons** hit the target, X-rays are produced (Fig 1.1e).

Heavy metal is a metal with high density, e.g. copper, lead and zinc.

When the electrons hit the heavy metal target, most of their KE is converted into the internal energy of the target. Less than 1% becomes the energy of X-rays.

Since the electrons are emitted from cathode, they are also called cathode ray.



Fig 1.1d *X-ray tubes.*

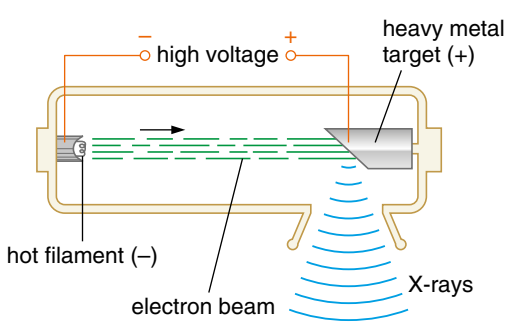


Fig 1.1e *Production of X-rays.*

Know more

Wilhelm Röntgen (1845–1923)

This is the first X-ray image of a human body part. It shows the hand of Röntgen's wife. Read the story of Röntgen to see how he discovered X-rays.

▸ p.42



Wilhelm Röntgen 倫琴

b Properties

X-rays are short-wavelength, high-energy electromagnetic radiation. They have high penetrating power and can penetrate through many substances, such as human flesh and organs.

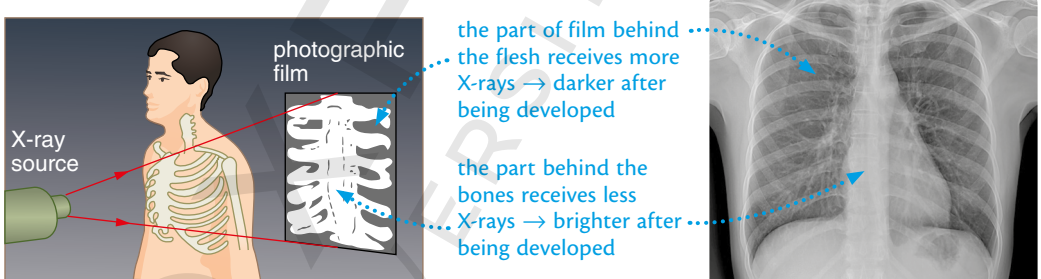
X-rays can produce an image on a **photographic film**. When the film is developed, parts receiving more X-rays appear darker than other parts receiving less X-rays.

They are ionizing and can damage living cells deep in the body. Long-term or strong exposure is hazardous to health.

c Applications

Low-frequency X-rays can penetrate flesh but not bones. High frequency X-rays can penetrate thin metals. By detecting the amount of X-rays penetrating different parts of an object, the internal structure of the object can be revealed (Fig 1.1f).

Cross-link
You may refer to Book E4 Ch 3 for more details about X-ray imaging. ▶



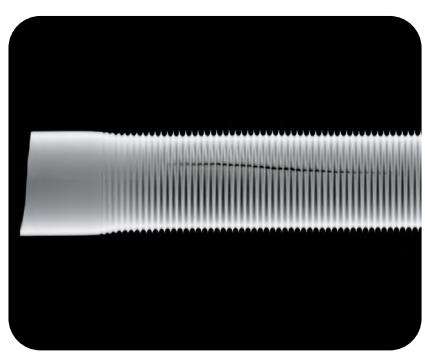
examine the inside of the body



analyze a thousand-year-old mummy



check travellers' luggage in airport



detect cracks within a metal tube

Fig 1.1f *Using X-rays to reveal internal structures of different objects.*

X-rays are EM waves of short wavelengths. They are produced by hitting heavy metals with fast electrons. They are ionizing and dangerous. They carry high energy and have high penetrating power, and can be used to reveal internal structures of objects.

photographic film 感光底片

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Are the following radiations ionizing? Put a tick in the corresponding box(es).

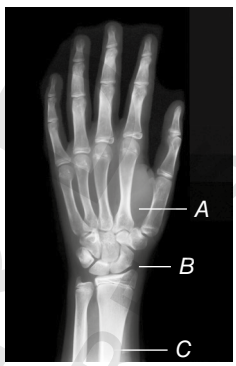
- ☐ Visible light from the screen of a phone
- ☐ X-rays used in medical diagnosis
- ☐ Microwaves used to heat food

2 *True or false:*

(a) X-rays are produced when fast electrons hit a heavy metal. (T/F)

(b) The more X-rays fall on a photographic film, the whiter the film when it is developed. (T/F)

3 In the X-ray image below, which part, *A*, *B* or *C*, received the least amount of X-rays?



Exercise 1.1

1 X-rays are produced when

A electrons are emitted from a hot filament.

B electrons move very fast.

C fast electrons hit a heavy metal target.

D a heavy metal heats up.

★**2** Which of the following statements about X-rays are correct?

(1) They have higher frequencies than microwaves.

(2) They can remove electrons out of molecules.

(3) They can penetrate flesh.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**3** Which of the following statements is/are correct?

(1) All EM waves are ionizing.

(2) All ionizing radiations are dangerous.

(3) X-rays are ionizing.

A (1) only

B (3) only

C (2) and (3) only

D (1), (2) and (3)

4 The simplified structure of an X-ray tube is shown below (Fig a).

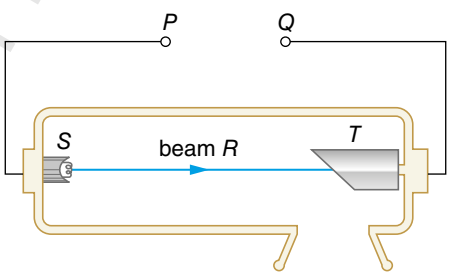


Fig a

(a) State the polarities of *P* and *Q*.

(b) State the direction of electric field between *S* and *T*.

(c) What does beam *R* consist of?

★**5** Figure b shows an X-ray image of a remote control.

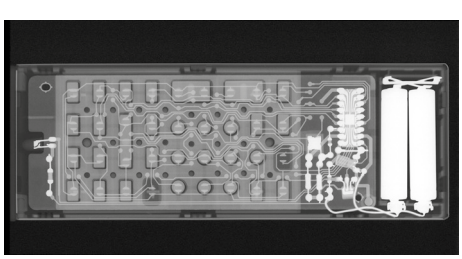


Fig b

(a) To obtain this image, what can be used to detect the X-rays passing through the remote control?

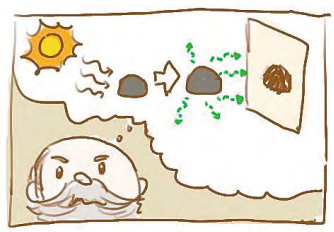
(b) State an application of X-rays.

(c) Suggest why such an image cannot be formed by using visible light.

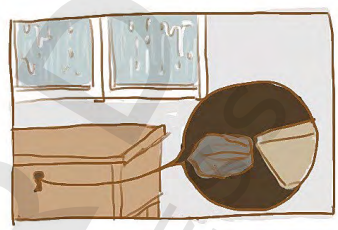
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1.2 Nuclear radiation

In 1896, *Henri Becquerel*, by accident, discovered that *uranium* emitted a new kind of radiation that could darken a photographic film (Fig 1.2a).



Becquerel first thought uranium salt emitted X-rays only after absorbing sunlight.



On a rainy day, he left some uranium salt with a photographic film together in a drawer.



After developing the film, he was surprised — the film registered unknown penetrating radiation!



He, together with *Marie* (his doctoral student) and *Pierre Curie*, were awarded the Nobel Prize in 1903.

Fig 1.2a *The discovery of radiation emitted by uranium.*

The radiation was **ionizing** like X-rays, however, also showed other different properties. Further studies found that the radiation is emitted by unstable nuclei. This kind of radiation is therefore called **nuclear radiation**. Unlike X-rays, nuclear radiation is emitted by a nucleus *spontaneously*, without any external stimulus (Fig 1.2b).

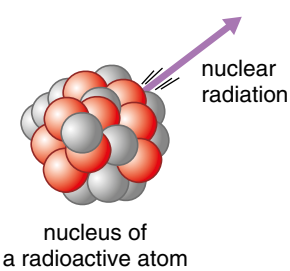


Fig 1.2b *Radioactivity of an unstable nucleus.*

The unstable atoms that emit nuclear radiation, like those in uranium salts, are **radioactive** and the phenomenon is called **radioactivity**.

Video

→ This video demonstrates an experiment simulating Becquerel's experiment.

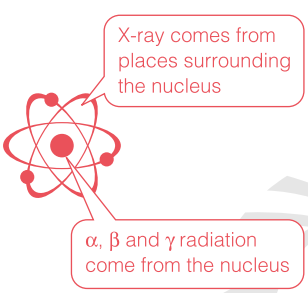
Know more

Henri Becquerel (1852-1908)

See more detail about Henri Becquerel.

▸ p.42

Ts may elaborate on the location within an atom that X-ray and nuclear radiation occur, so as to emphasize that the latter comes from the nucleus of an atom.



The term 'radioactivity' is coined by Marie Curie. Watch this video to know more about this great pioneer in nuclear science.

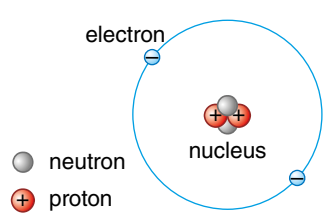
Useful link

Pre-study **Atomic structure**

An atom consists of a nucleus surrounded by electrons, and the nucleus consists of protons and neutrons.

Electrons are negatively charged. Protons are positively charged. Neutrons have no charge.

We will further discuss the atomic structure in Chapter 2.



Henri Becquerel 貝克勒耳　uranium 鈾　Marie and Pierre Curie 居里夫婦
nuclear radiation 核輻射　spontaneously 自發　radioactive 放射性　radioactivity 放射現象

DSE goal

Describe the origin and nature of α, β and γ radiations.

DSE exam

16(1A)Q32, 19(1A)Q31, 21(1A)Q33

1 Nature of nuclear radiation

There are three common types of nuclear radiation, namely **alpha** (α), **beta** (β) and **gamma** (γ). Table 1.2a gives a comparison of their natures and properties.

	α radiation	β radiation	γ radiation
Nature	streams of positively charged helium nuclei 	streams of negatively charged fast-moving electrons 	EM radiation of very high frequencies, similar to X-rays
Relative charge	+2 (each particle)	−1 (each particle)	no charge
Relative mass	~7300 (each particle)	1 (each particle)	no mass
Speed	a range of speeds: up to 10% of speed of light	a range of speeds: up to 90% of speed of light	a single speed: speed of light

The mass of a helium nuclei is 6.64×10^{-27} kg while the mass of an electron is 9.11×10^{-31} kg.

Table 1.2a *A comparison between three types of nuclear radiation.*

Nuclear radiation is the high-energy radiation emitted spontaneously by unstable nuclei. The three common types of nuclear radiation are α, β and γ.

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A radioactive nucleus emits nuclear radiation spontaneously. This means that the nucleus emits radiation

A suddenly by itself.

B when it is heated.

C when it is collided by a fast-moving electron.

2 Name the type of nuclear radiation (α, β or γ) corresponding to each of the following descriptions.

(a) Electrically neutral

(b) A stream of helium nuclei

(c) A stream of electrons

3 Categorize the following radiations. Put a tick in the corresponding box(es).

Radiation	EM radiation	Nuclear radiation	Ionizing radiation
visible light			
X-ray			
α radiation			
β radiation			
γ radiation			

alpha radiation α 輻射 beta radiation β 輻射 gamma radiation γ 輻射

The dawn of nuclear physics

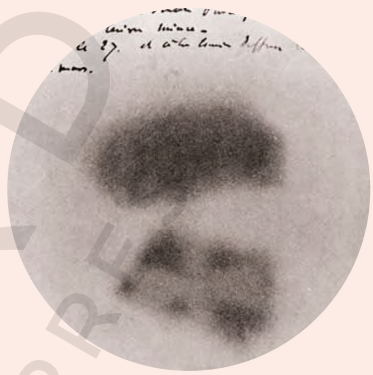
The discovery of nuclear radiation marked the start of a new branch of physics which studies the nucleus of an atom. Since Becquerel, a lot of effort has been put into understanding radioactivity. These studies eventually led scientists to understand the structure of the nucleus. Here are a few important milestones in the development of nuclear physics.

We will learn more about the nucleus in the coming chapters. As we will see, these studies also led to the development of some important modern technologies.

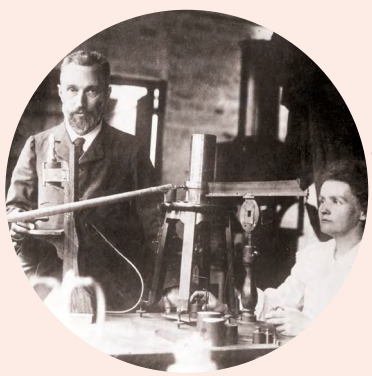
1895

1896

Henri Becquerel discovered the radiation emitted by uranium.



1905



1898

Marie and Pierre Curie discovered radioactive elements, polonium and radium.

1910

1900

Paul Ulrich Villard discovered gamma radiation.



1915

1911

Ernest Rutherford proposed that the atom has a nucleus.



1920



1925

1917

Ernest Rutherford discovered the proton.

1930

1932

James Chadwick discovered the neutron.



1934

Irène Curie and *Frédéric Joliot* created the first artificial radioactive element.

1935

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Paul Ulrich Villard 維拉德　Ernest Rutherford 盧瑟福　James Chadwick 查德威
Irène Curie 依倫娜居里　Frédéric Joliot 約里奧

2 Studying nuclear radiation in laboratory

We can learn more about nuclear radiation through experiments. Let us look at the radioactive sources and detectors used in school laboratory, and certain important safety rules.

a Radioactive source

In school laboratories, we use weak radioactive sources like *americium* and *radium* sealed in metal mounts (Fig 1.2c). The radioactive sources may emit more than one kind of nuclear radiation.

Radioactive sources sealed in metal mounts are called sealed sources.

Americium emits α and (a small amount of) γ radiation. Radium emits α, β and γ radiation.

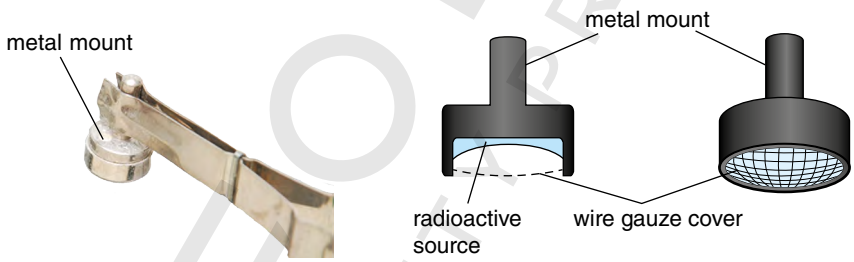


Fig 1.2c *Radioactive sources in school laboratories.*

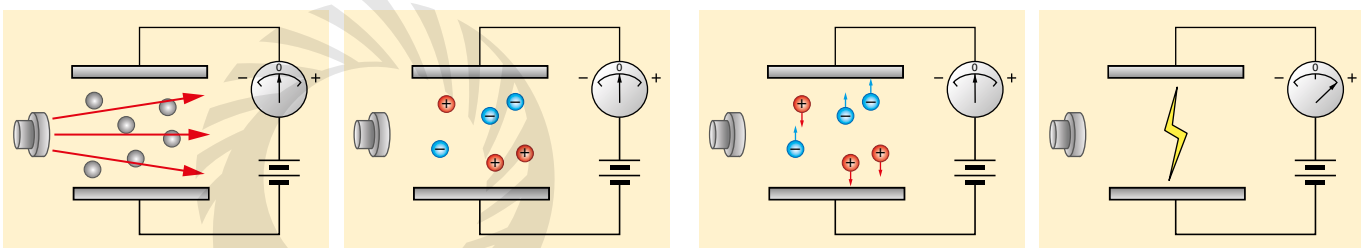
Meanwhile, ionizing radiation is always around us, even we do not introduce any radioactive sources deliberately. Such radiation is called **background radiation**. It comes from both natural and artificial sources.

Cross-link
You may refer to p.27 for details of background radiation.

b Detection of nuclear radiation

Many detectors detect the ions formed instead of measuring the radiation directly. When ionizing radiation passes through a gas, ions are formed. If the ions are formed between two electrodes under high voltage, current pulses will be produced (Fig 1.2d).

Caution
It is the ions, not the α or β particles, that produce the current pulses.



① Nuclear radiation ionizes gas molecules.

② Ions move towards the electrodes and produce current pulses.

Fig 1.2d *A gas becomes a conductor when ionized.*

The ions formed can also trigger other changes such as a change of state or chemical reaction. These changes are useful in the detection of nuclear radiation. Let us look at some detectors commonly used in a school laboratory.

americium 鎇　radium 鐳　background radiation 本底輻射

i Spark counter

A **spark counter** has an earthed wire gauze and a fine wire at a high voltage (about 4000 V) (Fig 1.2e). When a radioactive source is brought nearby, air is ionized and sparks are produced in the gap between the gauze and wire (Fig 1.2f). If more ions are produced, more sparks will be formed.

The mechanism of sparking is similar to that of lightning.

The spark counter is not an instrument used by professionals for detecting nuclear radiation. It is a school teaching apparatus. Strictly speaking, it is not a counter since it has no display of counts or count rate. However, we can get some idea of the 'count rate' from the rapidity of the sparks observed and the sparking sound heard.

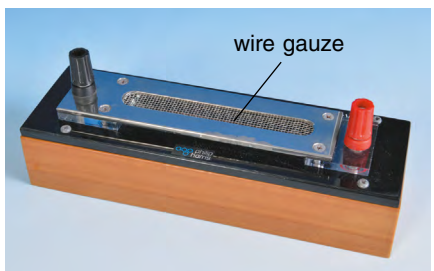


Fig 1.2e A spark counter.

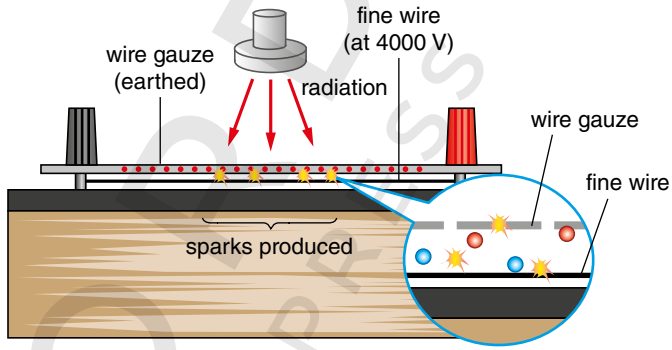


Fig 1.2f Sparks are produced in the gap between the wire gauze and the fine wire.

ii GM counter

A **Geiger-Müller counter**, or **GM counter** for short, consists of a GM tube connected to a counter (Fig 1.2g). When nuclear radiation enters the GM tube, it ionizes the gas and produces current pulses (Fig 1.2h). The counter shows the number of pulses it registered.



Fig 1.2g A Geiger-Müller counter.

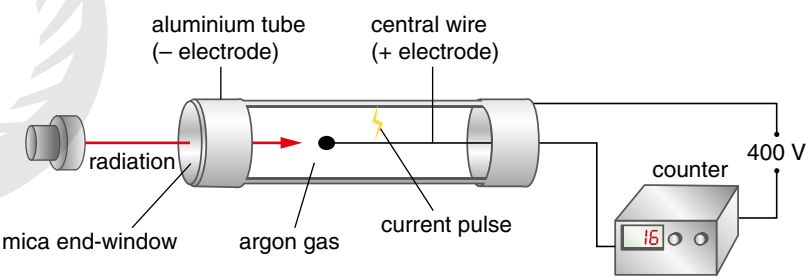


Fig 1.2h Working principle of a GM counter.

The number of pulses produced per unit time is called the **count rate**. If more ions are produced, the count rate will be higher.

spark counter 火花計數器　　Geiger-Müller counter (GM counter) 蓋革—彌勒計數器
count rate 計數率

✎ The diffusion cloud chamber is saturated with alcohol vapour which is cooled to an extremely low temperature by dry ice. A radioactive source inside the chamber emits radiation that produces ions along its path. The alcohol vapour condenses around the ions. The resulting tiny alcohol droplets show up as tracks in the bright light.

✎ It was said that C.T.R. Wilson, the inventor of the cloud chamber, suddenly got the insight from the formation of clouds while hiking in the Scottish mountains. He immediately cut short his holiday and returned to his laboratory to carry out experiments. He received the Nobel Prize in Physics in 1927 for his work on cloud chamber.

iii Diffusion cloud chamber

Although nuclear radiation is invisible, we can use a **diffusion cloud chamber** (Fig 1.2i) to visualize its tracks. The chamber is filled with vaporized alcohol. As radiation produces ions along its path, alcohol vapour condenses around the ions and white tracks are formed (Fig 1.2j). If more ions are produced, the tracks will be clearer.

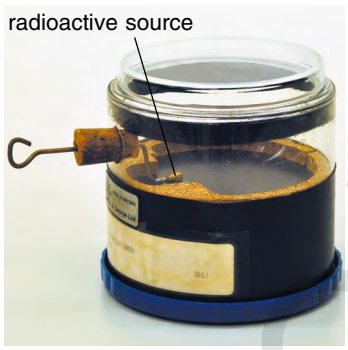


Fig 1.2i Diffusion cloud chamber.

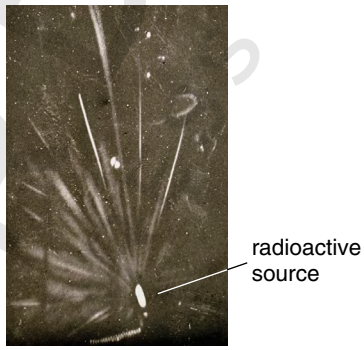


Fig 1.2j Some cloud chamber tracks.

iv Photographic film

A photographic film is a plastic plate coated with a material sensitive to radiation such as visible light and ionizing radiation. If the film is exposed to radiation, it appears blackened after being developed. The darkness of a developed film reflects the amount of radiation the film received (Fig 1.2k).

The film is wrapped to block visible light. ▶

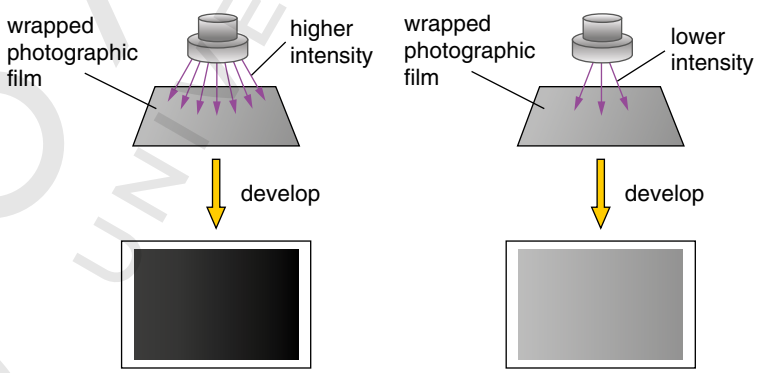
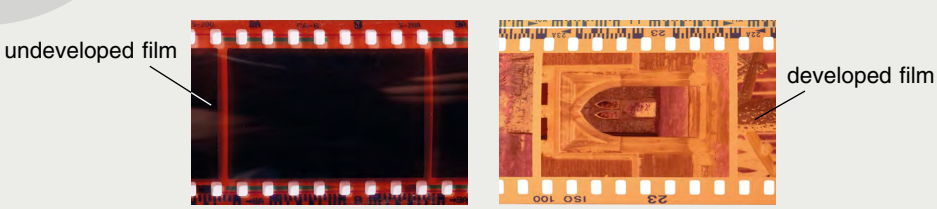


Fig 1.2k A photographic film becomes darker if it had a higher level of exposure to radiation.

Know more

Photographic development

Photographic development is a chemical process to transform the invisible latent image into a visible image.



diffusion cloud chamber 擴散雲室

DSE goal

Suggest safety precautions in handling radioactive sources.

C Laboratory safety

Nuclear radiation is very dangerous. Safety regulations regarding the use and the storage of radioactive substances should be strictly observed. Safety precautions are taken to reduce the risk of harm when handling radioactive sources. These include:

- Store all sources in a suitable lead container (Fig 1.2l). Keep radioactive sources locked in the containers when not in use.
- Handle the source with long forceps or special lifting tools. Take great care not to drop the source (Fig 1.2m).

✎ Ts may read the *Code of Practice on the Use of Sealed Radioactive Sources for Teaching Purposes in Schools* at http://www.rbhk.org.hk/docs/Code-of-Practice-e-201209.pdf.



Fig 1.2l *A container for storing radioactive source.*

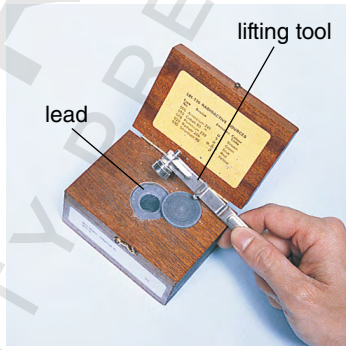


Fig 1.2m *Handle a radioactive source with a lifting tool.*



Fig 1.2n *The warning sign of radioactive sources.*

- Keep the radiation source at arm's length, and point it away from the human body, especially the eyes.
- Carefully plan the experiment to minimize the time of using the source. Have a practice run in advance.
- Label the storage place with warning signs (Fig 1.2n).

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The ionizing radiation present in the environment around us is called _____.

2 Among a diffusion cloud chamber, a GM counter and a photographic film, which one

(a) visualizes the path of invisible ionizing radiation;

(b) gives quantitative measurement of ionizing radiation?

3 *True or false*:

(a) If a weak radioactive source is dropped onto the floor, the best measure is to pick it up with bare hands as soon as possible. (T/F)

(b) Liquid radioactive wastes should be poured down the drain after an experiment. (T/F)

DSE goal

Compare α, β and γ in terms of penetrating power, ranges, ionizing power, behaviour in electric field and magnetic field, and cloud chamber tracks.

Detect radiation in terms of count rate using a GM counter.

Detect radiation with a photographic film and GM counter.

DSE exam

14(1A)Q32, 15(1A)Q32, 17(1A)Q31, Q32, 20(1A)Q30, 23(1B)Q9(c)(i)

Video

→ This video demonstrates Expt 1a.

Beware when handling the radioactive source. Read the safety precautions on p.12 thoroughly.

- Do not handle electrical appliances when your hands, feet or bodies are wet or when standing on a wet floor.
- The floor should be covered with effective insulating materials when conducting this experiment.
- Always connect the negative terminal to the earth terminal of an EHT power supply unless otherwise instructed.
- Always switch off the EHT power supply before making adjustments to the set-up.

3 Properties of nuclear radiation

How do the properties of α, β and γ radiations differ? In the following, we shall use the mentioned detectors to investigate their properties.

a Ionizing power

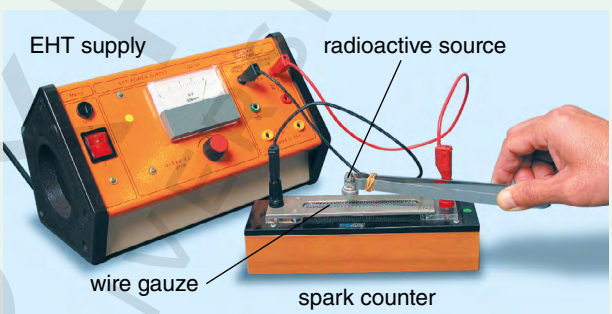
Ionizing power describes how well a radiation ionizes a material. To compare the ionizing power of the three types of nuclear radiations, we can use a spark counter to carry out the following experiment.

Experiment 1a / Ionizing power (for demonstration only)

1 Connect a spark counter to an EHT power supply. Set the output voltage of the power supply until the counter just stop sparking.

2 Hold an α source about 1 cm above the wire gauze of the spark counter. Count the number of sparks produced in one minute.

3 Repeat the experiment with a β source and a γ source in turn.



Discussion

Which type of radiation produces sparks at the highest rate? Which produces at the lowest rate?

The rate of sparking depends on the activity of the source and the type of radiation emitted by the source.

In Experiment 1a, the α source produces a lot of sparks, the β source produces a few and the γ source hardly produces any. The results can be concluded as follows (Fig 1.2o):

Ionizing power of the nuclear radiation:
α > β > γ

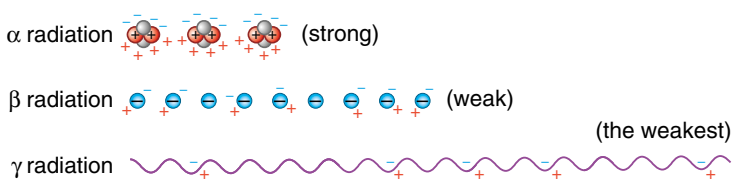


Fig 1.2o The ionizing power of α, β and γ radiations.

ionizing power 致電離能力

b Range in air

✎ Ts may explain the range of radiation by their ionizing power:
less ionizing power ⇒ lose less energy while moving through a medium ⇒ able to move over a longer distance

When radiation travels in a material and ionizes molecules, it loses energy. We say that it is gradually absorbed. The maximum distance a radiation travels in a material before it is **completely absorbed** is called its **range** in the material.

The range depends on the radiation type and the material. A strongly ionizing radiation has a short range.

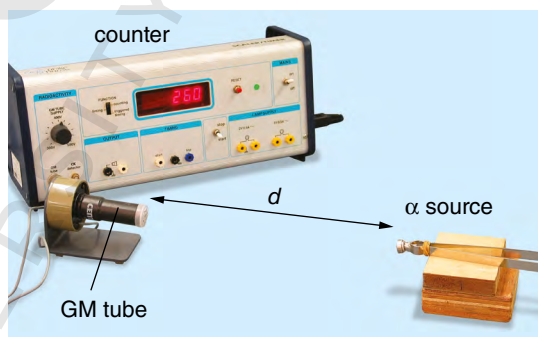
Range of the nuclear radiation in air:
γ (over 100 m) > β (several m) > α (a few cm)

Video

→ This video shows how a GM counter is used to monitor nuclear radiation.

Example 1 | Range of α radiation in air

The figure shows the set-up used to determine the range of α radiation in air.



Before the α source is introduced, the counter shows a count rate of around 20 min^{-1}. Then, the α source is brought near the GM tube. The distance d between them is measured by a ruler. As the source is moved away from the GM tube, the corresponding count rate R is recorded. The following table shows the result.

d / cm	0.5	1	1.5	2	2.5	3	3.5	4	4.5
R / min^{-1}	327	161	111	68	42	35	21	24	16

(a) Why does the GM counter record a non-zero count rate even when there is no α source in front of it?

(b) What is the count rate solely due to the α source when d = 2 cm?

(c) Estimate the range of this α source. Explain briefly.

Solution

(a) The non-zero count rate is due to background radiation.

(b) Count rate due to α source = 68 − 20 = 48 min^{-1}

(c) The range of the source is about 3.5 cm.
When the source is 3.5 cm away from the GM tube, R drops to about the same as the background count rate.

Cross-link
R fluctuates all the time. This is due to the random nature of radioactivity, which will be discussed in Ch 2. ▶

range 射程

→ This simulation is a 'virtual expt' on the penetrating power of nuclear radiation. Ss can find out about the reduction in count rate when each type of radiation is blocked by different absorbers. Also, Ss are asked to find out about some unknown sources from the reduction in count rate.

In the above example, the GM counter registers the count rate due to both the radioactive source and background radiation. If we only concern the radiation from the source, we need to obtain the **corrected count rate**.

Corrected count rate
= measured count rate – background count rate

C Penetrating power

Penetrating power describes how well a radiation penetrates matter. In the following experiment, we shall compare the penetrating power of the three types of nuclear radiation.

→ This video demonstrates Expt 1b.

Simulation Video

Experiment 1b / Penetrating power (for demonstration only)

1 Set up the GM counter. Without any source nearby, record the background count rate B.

2 Put an α source at about 1 cm from the GM tube. Record the count rate.

3 Insert sheets of paper one by one between the GM tube and the source until the count rate drops to a similar level as B. Record the number of sheets of paper.

4 Repeat step 3 with aluminium sheets and lead sheets.

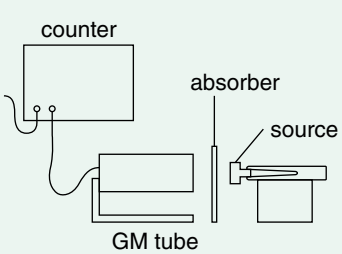
5 Repeat the experiment with a β source and a γ source in turn. You may put these sources at around 5 cm away from the GM tube.

Why should we put the α source so close to the GM tube? ▶

Beware when handling the radioactive source. Read the safety precautions on p.12 thoroughly.

✎ Note that the GM tube used in school is not very sensitive to α radiation since its mica end-window stops most of α radiation.





Discussion

1 Which absorbers completely block α and β radiation?

2 Do the absorbers completely block γ radiation?

penetrating power 穿透能力

From Experiment 1b, we find that (Fig 1.2p):

1 α radiation can be stopped by 1-2 sheets of paper.

2 β radiation can be stopped by 5 mm of aluminium.

3 γ radiation is absorbed by half after passing through 25 mm of lead. It cannot be stopped completely.

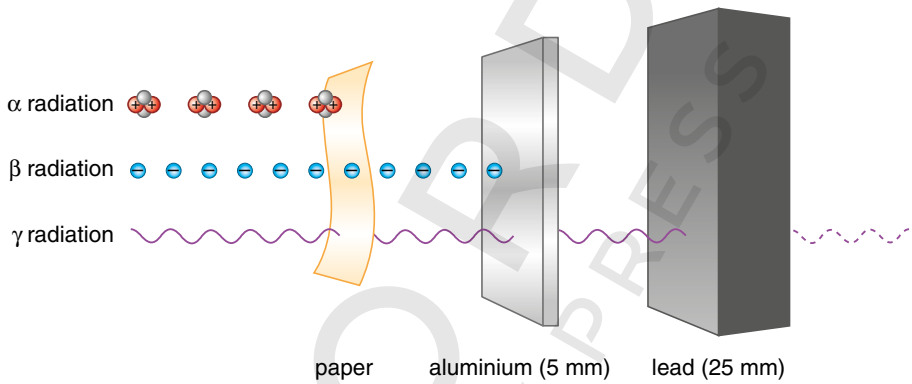


Fig 1.2p *The penetrating power of α, β and γ radiations.*

If a radiation produces more ions along its path, it loses energy faster and becomes less likely to penetrate through the matter. This results in a lower penetrating power. In general, a radiation with a strong ionizing power has a low penetrating power.

Penetrating power of the nuclear radiation:
γ > β > α

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

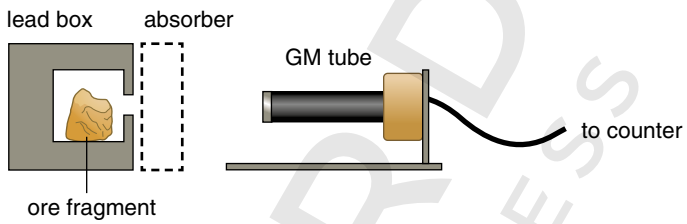
1 Arrange α, β and γ radiation in
(a) descending order of ionizing power;
(b) ascending order of range in air;
(c) ascending order of penetrating power.

2 In the following table, put a tick in the box if the absorber can stop the radiation **completely**.

Radiation	Absorber		
	5 pieces of paper	5-mm aluminium plate	25-mm lead block
α			
β			
γ			

Example 2 Distinguish radiation using different absorbers

A small fragment of radioactive ore is put in a lead box with a small opening. A GM tube connected to a counter is put in front of the opening.



Different absorbers are put between the ore fragment and the GM tube in turn and the corresponding count rates are recorded below.

Absorber	Count rate / min^{-1}
—	302
A sheet of paper	298
5-mm aluminium plate	33
25-mm lead block	29
50-mm lead block	32

(a) Explain whether the ore emits

(i) α radiation;

(ii) β radiation;

(iii) γ radiation.

(b) Estimate the background count rate. Explain briefly.

Solution

(a) (i) A piece of paper can stop α radiation. Since the count rate does not drop significantly when the paper is used as the absorber, the ore does not emit α radiation.

(ii) A 5-mm aluminium plate can stop β radiation. Since the count rate shows a significant drop when the aluminium block is used as the absorber, the ore emits β radiation.

(iii) Lead blocks can absorb some γ radiation. Since the count rate does not show **further** significant drop when lead blocks of different thicknesses replace the aluminium block, the ore does not emit γ radiation.

(b) A lead block can absorb all β radiation. When a 50-mm lead block is used, the remaining count rate is due to the background radiation. The background count rate is around 30 counts per minute.

▶ Exercise 1.2 Q10 (p.26)

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Radiations and absorbers

① Identify the type of radiation that can be absorbed by each absorber.

② Start from the weakest absorber.

③ Compare the count rates. Since an absorber may be able to absorb more than one type of radiation, you need to compare the counts rates between cases of different absorbers, not just with the case with no absorber.

Example 3 | Film badge

Radiation workers need to wear film badges to monitor their exposure to radiation during work (Fig a). The film badge consists of a 5-mm aluminium slab, a 5-mm lead slab and a photographic film in a plastic wrapping (Fig b). The film is developed in regular time intervals. The degree of blackening of different areas of the film shows the level of exposure to β and γ radiation.

The plastic wrapping protects the film from moist and visible light. ▶

→ This article provides more information about film badge.

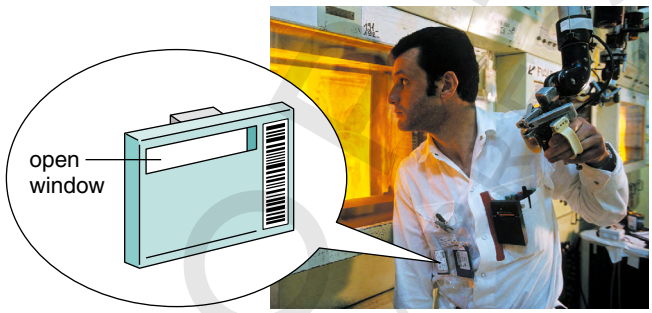


Fig a

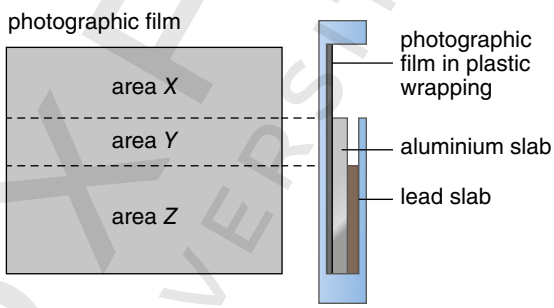


Fig b

In each of the following cases, compare and explain the degree of blackening of different areas of the film after it has been developed.

(a) The worker is exposed to β radiation only.

(b) The worker is exposed to γ radiation only.

Solution

(a) Area *X* is blackened because the plastic wrapping cannot block β radiation effectively. Areas *Y* and *Z* are not blackened because β radiation is completely blocked by the aluminium slab and lead slab.

(b) Areas *X* and *Y* show similar degree of blackening because the plastic wrapping and the aluminium slab cannot block γ radiation effectively. Area *Z* is the least blackened because the lead slab can partly block γ radiation. This area receives the least amount of radiation.

▶ Revision 1 Q18 (p.39)

→ Teaching notes and exercises are available on OUP web.

Right and wrong



✗ α radiation is ionizing. It can blacken a film and be detected by a film badge.

✓ Film badge cannot detect α radiation because the plastic wrapping of the film completely blocks α radiation.

Skill — Problem-solving

Distinguishing radiations using different absorbers

We can draw a diagram to help organize our thoughts.

For β only:
- can reach area *X*
- cannot reach area *Y* & *Z*

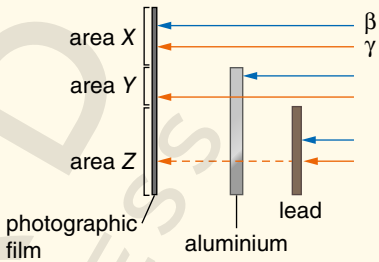
⇒ only *X* is blackened

For γ only:
- can reach area *X* & *Y*
- some can reach area *Z*

⇒ blackening in *X* ~ *Y* > *Z*

For β and γ:
- both can reach area *X*
- only γ can reach area *Y*
- only some γ can reach area *Z*

⇒ blackening in *X* > *Y* > *Z*



β	γ	β + γ
X	X	X
Y	Y	Y
Z	Z	Z

d Cloud chamber tracks

Sometimes, seeing is believing. In the following experiment we shall see how the tracks of α, β and γ radiations look like.

→ This video demonstrates Expt 1c.

Video

Be careful when handling the radioactive source. Read the safety precautions on p.12 thoroughly.

Do **not** touch the cloud chamber and dry ice with bare hands.

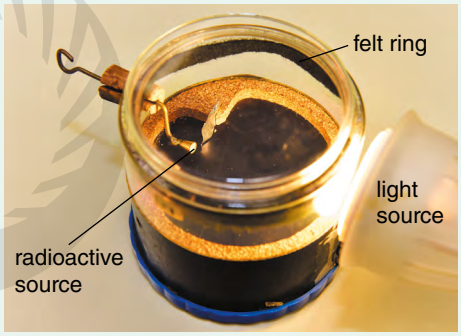
Experiment 1c — Diffusion cloud chamber tracks (for demonstration only)

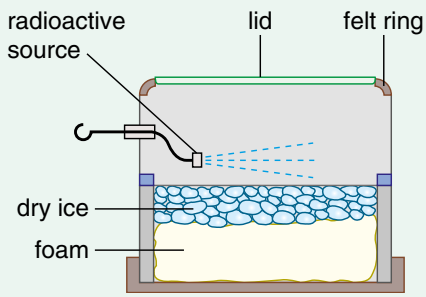
1 Put some dry ice in the lower part of the cloud chamber.

2 Add some alcohol to the felt ring. Cover the chamber with the lid.

3 Illuminate the chamber with a lamp.

4 Insert different types of radioactive sources in turn and observe from above the tracks produced. Sketch the tracks observed.





Discussion

Describe the tracks observed. What are the differences between the tracks formed by different types of radiation?

You may see cloud chamber tracks of other particles on this website.

Useful link

The characteristics of the tracks are closely related to the ionizing power and the nature of the radiation. Table 1.2b compares the three kinds of tracks.

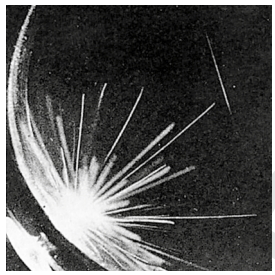
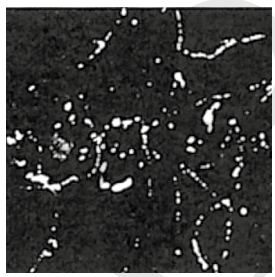
	Cloud chamber tracks	Interpretation
α radiation	thick and straight 	• Thick tracks: α particles produce a large number of ions along their paths. ⇒ α radiation has a strong ionizing power. • Straight tracks: α particles are not deflected easily in collisions. ⇒ α particles are heavy.
β radiation	thin and twisted 	• Thin tracks: β particles produce a small number of ions along their paths. ⇒ β radiation has a weak ionizing power. • Twisted tracks: β particles are easily deflected in collisions. ⇒ β particles are light.
γ radiation	scattered and hardly visible 	• γ radiation has a very weak ionizing power.

Table 1.2b *The tracks of different nuclear radiations in a cloud chamber.*

→ This video demonstrates the formation of right-angled fork tracks.

Video

In the early 20th century, cloud chambers were extensively used for investigating subatomic particles. Scientists found that α radiation produced **right-angled fork tracks** in a cloud chamber filled with helium (Fig 1.2q). These tracks are formed when the two colliding particles have the same mass. This supports the view that α radiation is a stream of helium nuclei.

For two particles of the same mass, the angle between the direction of motions after collision is always 90° due to the conservation of energy and momentum.

The angle of the fork is slightly less than 90° because the track was photographed at an angle.

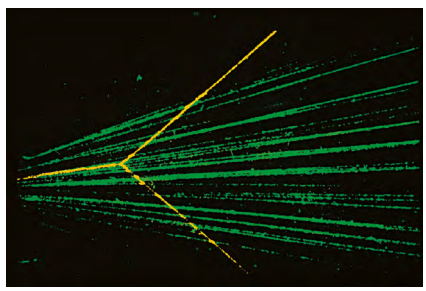


Fig 1.2q *Right-angled fork track.*

right-angled fork track 直角分叉徑跡

e Deflections in electric and magnetic fields

As we know, α and β radiations are streams of charged particles. When they travel in an electric field or a magnetic field, they are deflected by the electric force or magnetic force acting on them respectively (Fig 1.2r). γ radiation has no charge and is not affected by the fields.

In fact, this is how scientists discovered that α and β radiations carry electric charges.

Ts may need to refresh Ss' memory about the electric force and magnetic force.

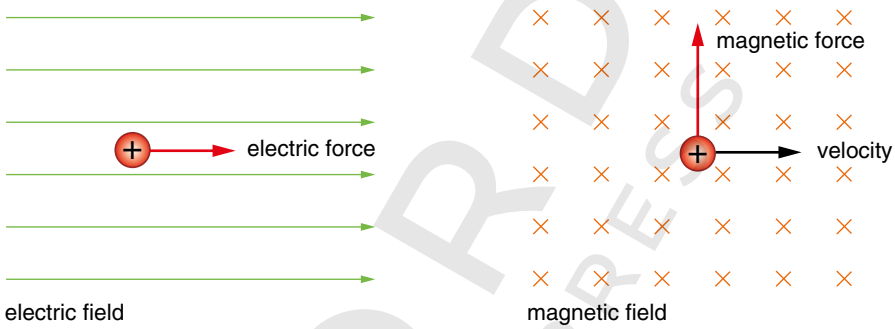


Fig 1.2r Forces experienced by a positively charged particle in an electric field and a magnetic field.

Note that the mass of an α particle is about 7300 times that of a β particle, so **α radiation is deflected by a much smaller amount** in both electric and magnetic fields.

i Deflections in an electric field

In an electric field (Fig 1.2s),

- α radiation (+) is deflected towards the negative side;
- β radiation (–) is deflected towards the positive side;
- γ radiation (neutral) is not deflected.

The lead sheets ensure that the radiation entering the field is unidirectional.

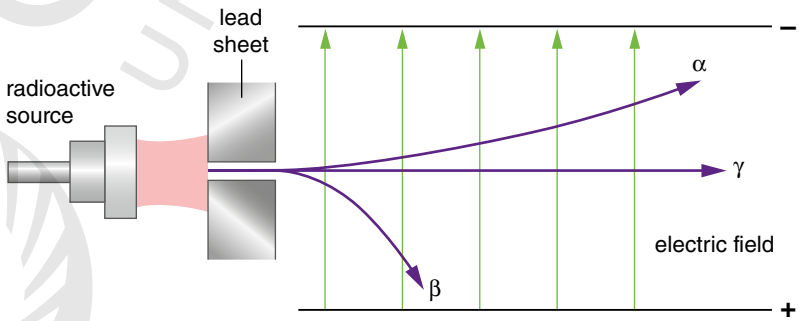


*Fig 1.2s Deflections of α, β and γ radiations in an electric field (deflections **not** to scale).*

Right and wrong

✗ The quantity of charge of an α particle is double that of a β particle, so α radiation has a larger deflection in an electric field.

✓ The mass to charge ratio of an α particle is much larger than that of a β particle, so α radiation has a much smaller deflection in an electric field.

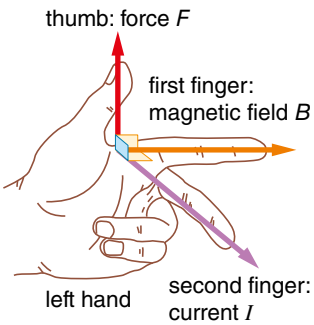
→ This simulation demonstrates the deflection of α, β and γ radiations in a magnetic field.

Simulation

ii Deflections in a magnetic field

In a magnetic field, the directions of deflection of α and β radiations can be determined by using Fleming's left-hand rule (Fig 1.2t). γ radiation is not deflected.

Fleming's left-hand rule:



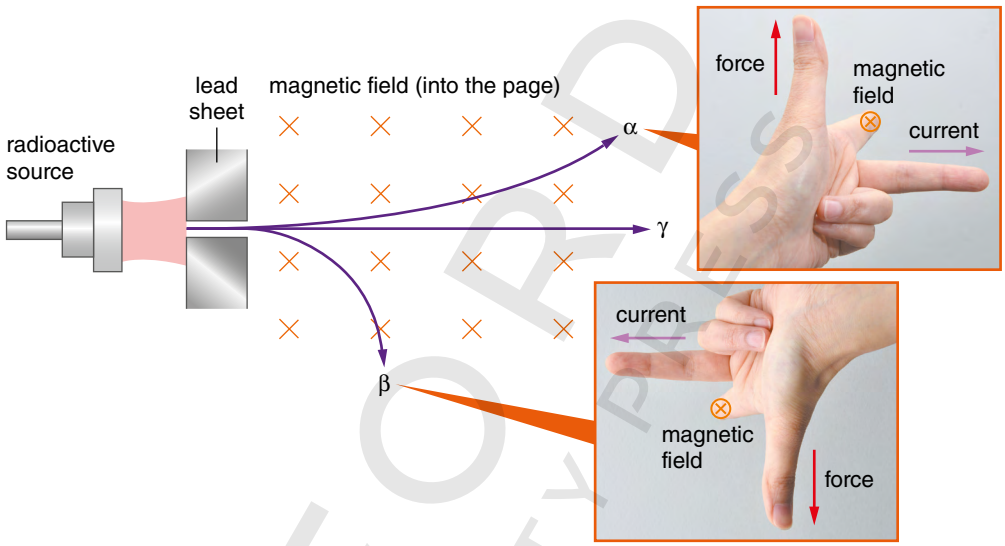


Fig 1.2t *Deflections of α, β and γ radiations in a magnetic field (deflections* ***not*** *to scale).*

We may verify the deflection of β radiation by a magnetic field in the following experiment.

✎ Ts may need to mention the relation between direction of current and direction of travel of α and β particles by considering the type of charge they are carrying.

→ This video demonstrates Expt 1d.

Video

⚠ Be careful when handling the radioactive source. Read the safety precautions on p.12 thoroughly.

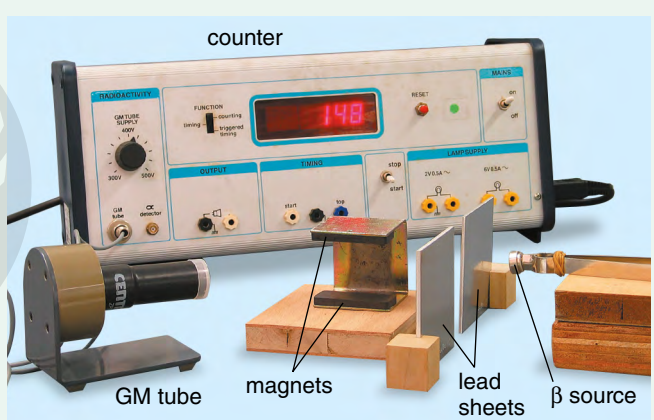
✎ β particles are emitted at a range of speeds. They undergo different degree of deflection.

Experiment 1d Magnetic deflection of β radiation (for demonstration only)

1 Set up the apparatus. Align the β source with the GM tube. Measure the count rate without the magnets.

2 Put the magnets in place. Note the drop in the count rate.

3 Move the GM tube to different positions and note the count rate.



Discussion

In which direction does the β radiation deflect? Does this agree with your prediction?

✎ The answer depends on the direction of the magnetic field.

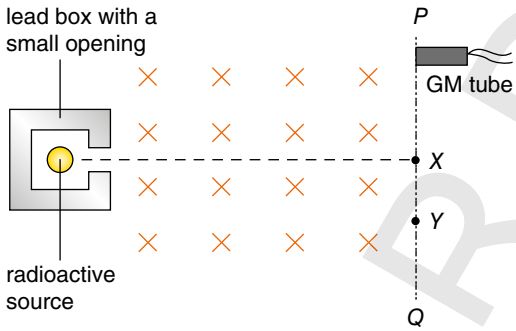
☆ HKDSE Practice Paper 2012 Paper 1A Q35

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Nuclear radiation in a magnetic field

A radioactive source is placed in front of a uniform magnetic field pointing into the page. When a GM tube is moved along *PQ*, maximum count rates are detected at *X* and *Y*.



What happens to the count rates at *X* and at *Y* when a stronger magnetic field is applied?

A Both increase.

B Both remain the same.

C The count rate at *X* remains the same and that at *Y* decreases.

D The count rate at *X* decreases and that at *Y* increases.

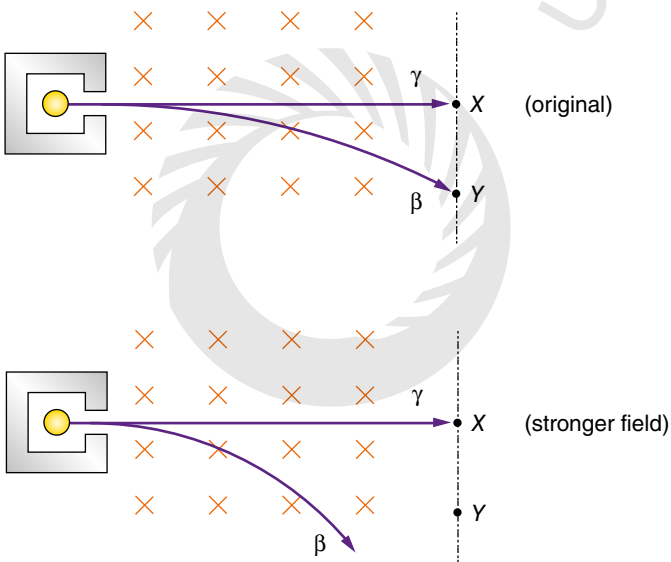
Solution

The radiation reaching *X* is γ radiation and that reaching *Y* is β radiation. When a stronger magnetic field is applied, β radiation will deflect more but γ radiation remains undeflected.

Therefore, the maximum count rate due to β radiation will be detected at a position below *Y*. This means the count rate at *Y* will decrease.

∴ The answer is C.

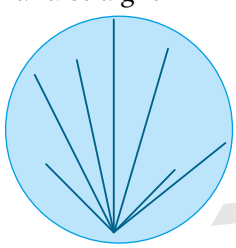
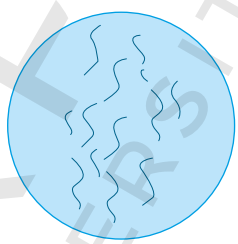
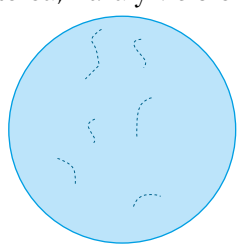
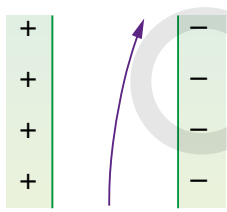
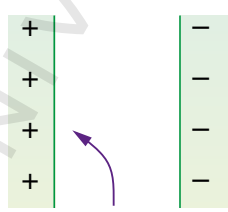
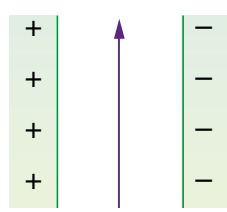
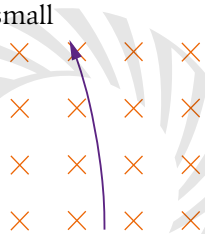
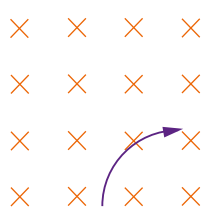
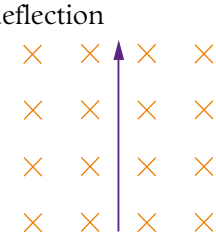
Common mistake ⚠
Students may wrongly think that a stronger magnetic field would make more radiation deflect. In fact, the strength of magnetic field affects only the degree of deflection.



▶ Revision 1 Q13 (p.38)

Summary of properties of nuclear radiation

Table 1.2c summarizes the nature and the major properties of α, β and γ radiations:

	α radiation	β radiation	γ radiation
Nature	streams of positively charged helium nuclei	streams of negatively charged fast-moving electrons	EM waves (very high frequency)
Relative charge	+2 (each particle)	–1 (each particle)	no charge
Relative mass	~7300 (each particle)	1 (each particle)	no mass
Speed	up to 10% of speed of light	up to 90% of speed of light	speed of light
Ionizing power	strong	weak	very weak
Penetrating power	low, stopped by 1–2 sheets of paper	high, stopped by 5-mm aluminium	very high, halved by 25-mm lead
Range in air	a few cm	several m	over 100 m
Cloud chamber track	thick and straight 	thin and twisted 	scattered, hardly visible 
Deflection in electric field	very small 	large 	no deflection 
Deflection in magnetic field	very small 	large 	no deflection 
Detectors [1] [2]	spark counter, GM counter, diffusion cloud chamber, photographic film	GM counter, diffusion cloud chamber, photographic film	GM counter, diffusion cloud chamber, photographic film

[1] *The ionizing power of β and γ radiations may not be strong enough. Sometimes they may not be detected by a spark counter.*

[2] *The paper or plastic wrapping of a photographic film blocks α radiation from blackening the film.*

Table 1.2c *The nature and properties of α, β and γ radiations.*

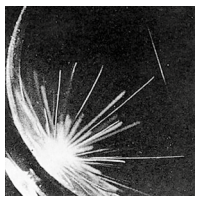
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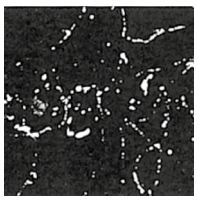
Quick check 5

→ E-version in OUP Exercise Platform is also available. Google Forms Microsoft Forms

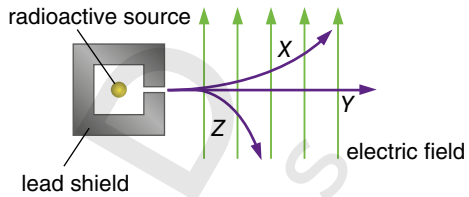
1 Name the radiation that produces the cloud chamber tracks below.

(a) 

(b) 

(c) 

2 A radioactive source emits α, β and γ radiation into an electric field.



(a) What are *X*, *Y* and *Z*?

(b) If the electric field is replaced by a magnetic field, what is the direction of the magnetic field so that *X* is still deflected upwards?

Exercise 1.2

1 Two radioactive sources *X* and *Y* produce different kinds of cloud chamber tracks (Fig a).

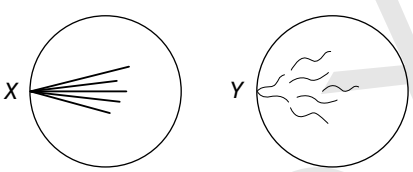


Fig a

What kind of radiation does each of sources *X* and *Y* emit?

	Source *X*	Source *Y*
A	α	β
B	α	γ
C	γ	β
D	β	γ

★**2** A manufacturer produces a tool that can remove static charges on objects by using ionizing radiation. The tool contains a radioactive source that emits radiation with a strong ionizing power. Which of the following sources is the most suitable for the tool?

A α source

B β source

C γ source

D The above sources are equally suitable.

★**3** Which of the following detectors can detect α radiation?

(1) Diffusion cloud chamber

(2) Photographic film in a plastic wrapping

(3) GM counter

A (1) only B (3) only

C (1) and (2) only D (1) and (3) only

★**4** A radioactive source produces some cloud chamber tracks in a very strong magnetic field (Fig b). What are the type of the radiation and the direction of the magnetic field?

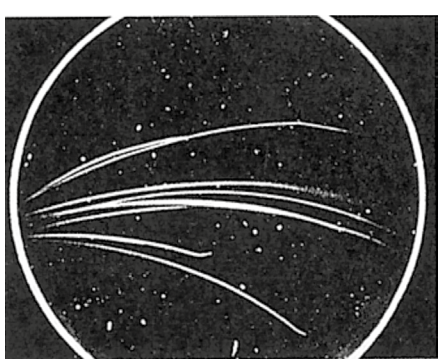


Fig b

	Type of radiation	Direction of magnetic field
A	α	into the page
B	α	out of the page
C	β	into the page
D	β	out of the page

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*5 An α particle and a β particle enter an electric field. Which of the following is correct?

A The magnitude of electric force acting on the α particle is equal to that on the β particle.

B The magnitude of electric force acting on the α particle is larger than that on the β particle.

C The magnitude of electric force acting on the α particle is smaller than that on the β particle.

D The magnitude of the electric force acting on the particle depends on the speed of the particle.

*6 Figure c shows an experiment similar to Becquerel's. A metal key is put on a wrapped photographic film. A β source is hovered above the key.

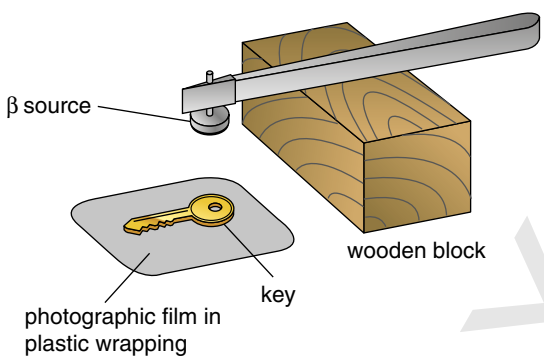


Fig c

The film is developed after half an hour. Which of the following correctly shows the image on the photographic film?

A 

B 

C 

D 

*7 Explain why the cloud chamber tracks of β radiation are thin and twisted.

*8 A radioactive source supposedly emits γ radiation only. Describe one method to check it does not emit α and β radiations.

*9 In Figure d, the radioactive source shown emits α, β and γ radiations into a magnetic field. Sketch the paths of the radiations and label them in the figure.

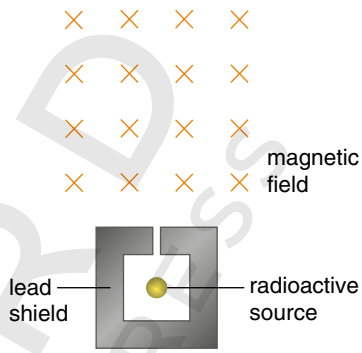


Fig d

*10 A GM counter is used to study a radioactive source (Fig e).

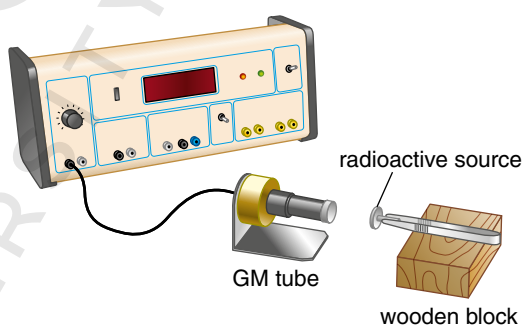


Fig e

The count rates recorded by the GM counter when the source is placed at different distances are recorded in Table a.

Distance / cm	Count rate / min^{-1}
2	653
5	129
10	27
100	31
300	29

Table a

(a) State an approximate count rate due to background radiation.

(b) Which type(s) of nuclear radiation does the source emit? Explain briefly.

(c) If the source is placed at a distance of 5 cm from the GM tube and a piece of paper is placed between them, predict the count rate.

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1.3 Radiation safety

DSE goal

Represent radiation equivalent dose using the unit sievert (Sv).

Discuss potential hazards of ionizing radiation and the ways to minimise the radiation dose absorbed.

Can you identify anything radioactive in Figure 1.3a? The answer may be surprising — the air, the plant, the monkey and even the banana it holds are all radioactive.

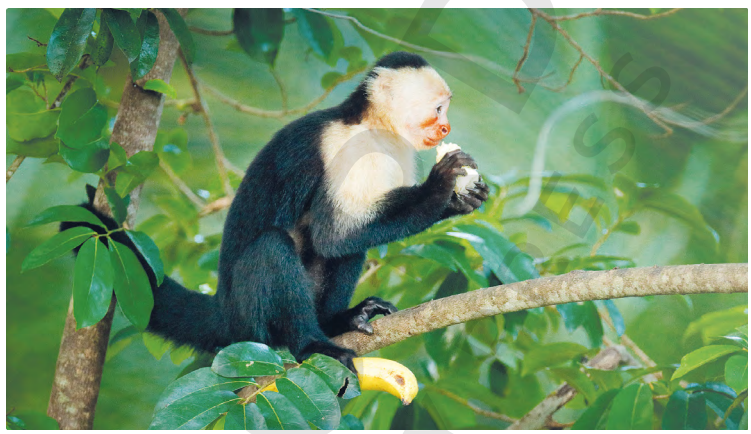


Fig 1.3a *A monkey eating banana on a tree.*

1 Background radiation

Background radiation mainly arises from natural sources, including cosmic rays coming from space and natural radioactive substances on the Earth (Fig 1.3b).



Some cosmic rays from outer space are ionizing.



Soil and rocks contain radioactive minerals. They also release a radioactive gas, namely radon.



Granite, a common building material in Hong Kong, releases radon into air.



Food and water contain a trace amount of radioactive substances.

Fig 1.3b *Natural sources of background radiation.*

It is important to have good ventilation in your house to prevent radon gas from accumulating. ▶

The radioactive substances produced in nuclear weapon tests in the last century can still be detected today.

Human activities may also produce radioactive fallout and wastes. They contribute a small part to the background radiation (Fig 1.3c).



Nuclear weapon explosions and testing produced radioactive fallout.



Radioactive discharge from nuclear power plants and fuel factories may enter the environment.

Fig 1.3c *Artificial sources of background radiation.*

We are affected by background radiation via direct irradiation, inhalation and ingestion of radioactive substances. Some radioactive substances are absorbed by our body; hence our bodies are in fact slightly radioactive.

2 Biological effect of ionizing radiation

Ionizing radiation is dangerous because it can produce highly reactive ions in cells. The ions may take part in chemical reactions and disrupt the normal functions of healthy cells, or even damage genetic material (Fig 1.3d).

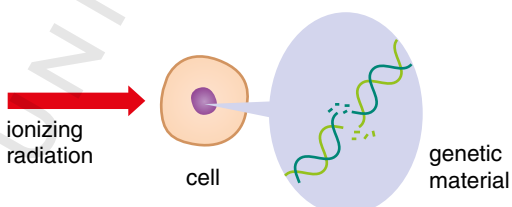


Fig 1.3d *Ionizing radiation may break genetic material in a cell.*

Some harmful effects are *deterministic* (can be predicted), while some are not (Table 1.3a).

	Deterministic effects	Non-deterministic effects
Nature	• occur when dose is above threshold • higher dose → more severe	• occur by chance • higher dose → higher chance
Example	• acute damage of organs, e.g. skin damage (Fig 1.3e) • latent effect, e.g. infertility, cataract	• cancer • genetic effect

Table 1.3a *Effects of ionizing radiation.*



Fig 1.3e *Skin damage by ionizing radiation.*

deterministic 確定的

When the radioactive source is **outside** the body, γ radiation is the most damaging as it can penetrate tissues deep in the body. β radiation can also penetrate our skin and damage the cells inside. α radiation can be blocked by our skin, causing little harm (Fig 1.3f(i)).

However, when a radioactive source is **inside** the body, α radiation becomes the most damaging as it has the strongest ionizing power. β and γ radiation are relatively less harmful than α radiation as they have weaker ionizing power (Fig 1.3f(ii)).

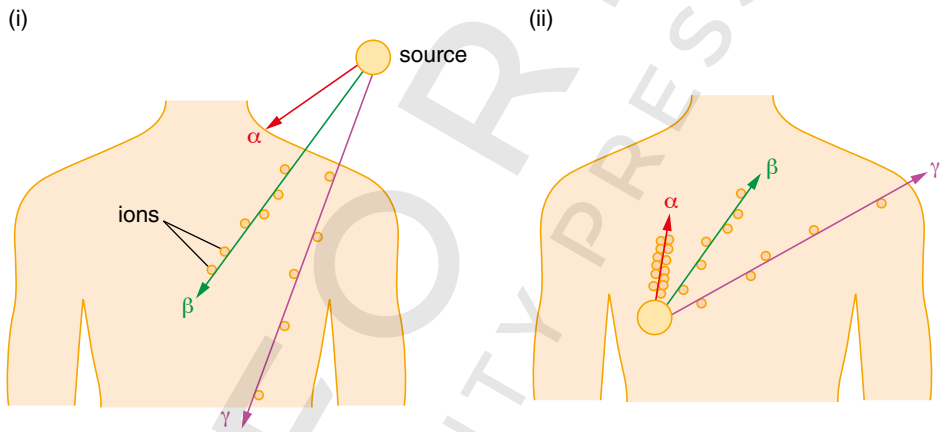


Fig 1.3f *A radioactive source (i) outside, (ii) inside the human body.*

Know more

Radioactive consumer products

When nuclear radiation was just discovered, people were not aware of its harmful effects. Consumer products, like cosmetics, toothpaste and jewelleries, were added with radium and other radioactive substances in the early 20th century.



Poster advertising radioactive toothpaste in early 20th century.

Radiation from cigarettes

Smoking is hazardous to health. One of the reasons is that cigarettes contain radioactive substances such as lead-210 (Pb-210) and polonium-210 (Po-210). Since Po-210 emits α radiation, it will cause great harm to the lungs when inhaled.



OUP eResources

✎ Po-210 undergoes α decay and has a half-life of 140 days. Pb-210 undergoes β decay and has a half-life of 22 years.

3 Equivalent dose

Our bodies can tolerate and repair a certain low level of damage caused by ionizing radiation. However, if the amount of radiation that we receive, or the radiation **dose**, is over a safety limit (i.e. an *overdose*), it will be more difficult to repair the damage.

The biological effect of ionizing radiation on the human body can be described by the **equivalent dose**, measured in **sieverts** (**Sv**). Figure 1.3g shows some effects of ionizing radiation at different dosages.

Different types of radiation differ in their biological effects on the human body. To reflect this difference, the absorbed dose is multiplied by a radiation weighting factor (α: 20, β: 1, γ: 1). The product is the equivalent dose. One sievert represents the equivalent biological effect of the absorption of one joule of radiation energy per kilogram of human tissue.

Some parts of the body are more easily damaged by radiation than others. To reflect this different sensitivity, the equivalent dose is multiplied by a tissue weighting factor (e.g. gonads: 0.2, lung and stomach: 0.12, skin: 0.01). Effective dose is the sum of the weighted equivalent doses in all the tissues and organs of the body, also measured in Sv.

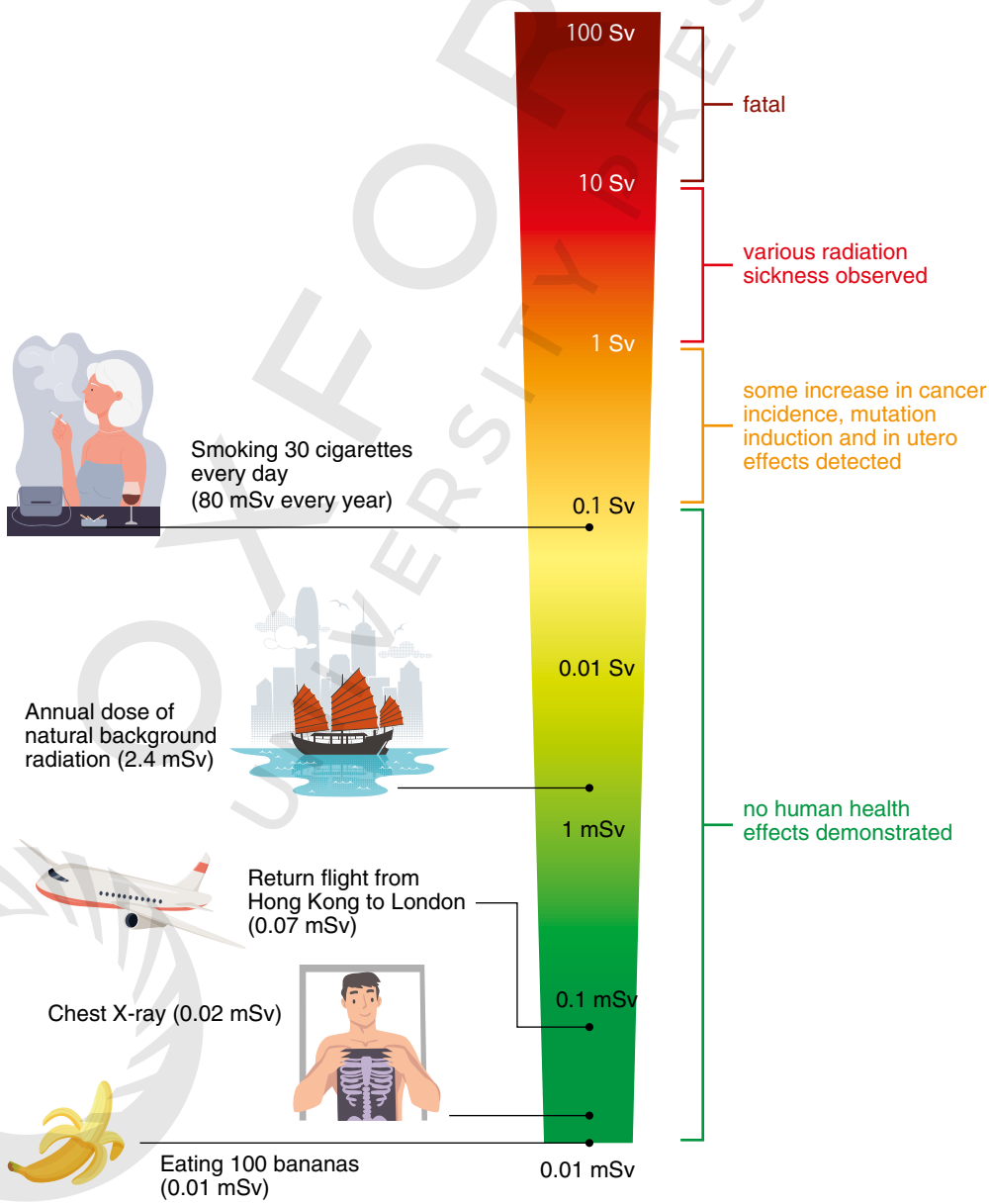


Fig 1.3g *Effects of ionizing radiation.*

Know more

Banana equivalent dose

As 1 Sv is a large dose, mSv and μSv are used more often in daily life. The equivalent dose of a banana is also used as an informal unit.

▸ p.41

The effects of radiation are accumulative. A low dose of radiation may not cause immediate effects; however, it may still be hazardous to our health if the dose is received for a long period of time.

dose 劑量　　overdose 過量　　equivalent dose 當量劑量　　sievert (Sv) 希沃特

✎ The dose may vary in different places.

The annual dose absorbed by an average person is about 3 mSv. About 80% comes from natural sources and the remaining 20% comes from artificial sources. Figure 1.3h shows the radiation doses from different sources of ionizing radiation.

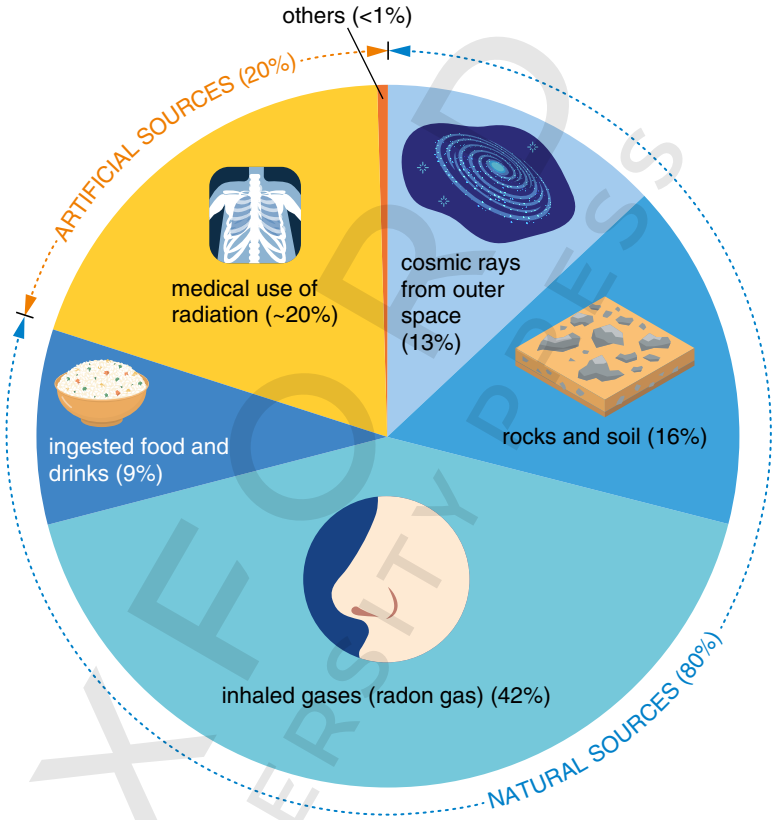


Fig 1.3h *Sources of radiation dose received by an average person.*

4 Protection against radiation

To protect ourselves from ionizing radiation, the rule of thumb is to reduce the exposure to radiation. Here are some safety principles:

1 **Time**: Minimize the exposure time to radiation.

2 **Distance**: Stay away from radioactive sources. The further away from the sources, the lower the dose received.

3 **Shielding**: Use barriers to shield radiation. Materials like lead, concrete and water give good protection from radiation (Fig 1.3i).

4 **Containment**: Make sure the radioactive sources are kept out of the environment.



Fig 1.3i *Protective clothing made of lead.*

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 What is the unit of equivalent dose?

A Ci B Sv

C Bq D J

2 *True or false:*

(a) Background radiation mainly comes from natural sources. (T/F)

(b) Ionizing radiation is dangerous because they can damage our living cells. (T/F)

(c) γ radiation is always the most dangerous nuclear radiation. (T/F)

3 Which of the following are valid methods to protect a person from ionizing radiation? Put a tick in the corresponding boxes.

☐ Store radioactive sources in a room with concrete walls.

☐ Cover body parts with lead shields.

☐ Keep a distance from radioactive sources when using them.

☐ Avoid unnecessary X-ray scans.

Exercise 1.3

1 Which type of radiation has the strongest biological effects on the human body when the source is inside the body?

A α radiation

B β radiation

C γ radiation

D X-ray

2 Which of the following is a natural source of ionizing radiation?

A Drinking water

B A dead animal

C A brick

D All of the above

3 Which of the following causes the highest radiation dose for an average person?

A Food

B Radon gas

C Cosmic radiation

D Medical X-rays

4 The equivalent doses of some medical procedure are listed in Table a.

Procedure	Equivalent dose / mSv
X-ray imaging, chest	0.1
X-ray imaging, abdominal	0.7
CT scan, head	2.1
CT scan, spine	6

Table a

Which of the following statements is correct?

A A CT scan of the head will definitely cause cancer.

B It is safe to have unlimited times of X-ray imaging of the chest.

C X-ray imaging and CT scans should be used only when necessary.

D Ionization will not occur in having one X-ray imaging of the abdominal.

★**5** Explain why we should take the following safety precautions when handling radioactive sources in a school laboratory.

(a) Keep the source at arm's length.

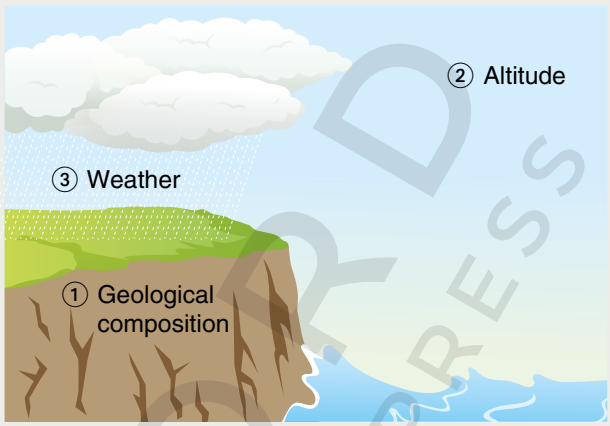
(b) Carefully plan the experiments to minimize the time that the source is used.

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Know more

Variations of background radiation level in Hong Kong

The level of background radiation differs from place to place and is affected by many factors.



① Geological composition: Soil and rocks contain more radioactive substances than seawater.

② Altitude: Cosmic rays are stronger at higher altitudes as they pass through less air.

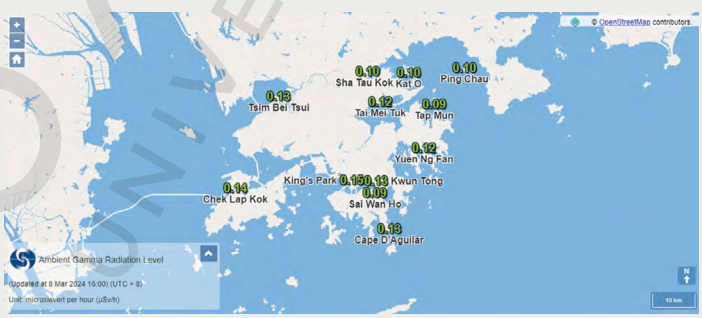
③ Weather: In rainy days, rain washes down natural radioactive substances to the ground.

Radiation monitoring and assessment in Hong Kong

The Hong Kong Observatory is responsible for monitoring the radiation level in Hong Kong. Samples of air, soil, food, water, etc. are collected and analysed to check if the radiation level is normal and unaffected by nuclear power plants nearby.

Useful link

The HKO has organized school-based workshop called 'Gamma-Go', in which Ss can measure and analyse γ radiation data collected at the local environment. For details please refer to HKO's website.



Real-time γ radiation level in air is available on the Observatory website.

STSE

Health check plans for better health?

Regular medical checkups can help us better understand our health condition. They can also increase the possibility of detecting certain diseases at an early stage and minimizing the threat of some critical diseases.

There are many health check plans on the market. Some of these plans include a wide range of tests. However, some of the tests such as chest X-rays make use of ionizing radiation, which may increase the risk of cancer. Is a plan with more tests always better than one with fewer tests? What should be considered before we decide to buy or not to buy a health check plan?

Key concepts 1

1 Ionizing radiation

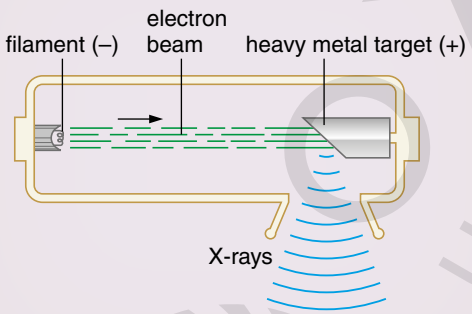
Radiation
Energy emitted by a source that travels through a medium or a vacuum

Non-ionizing radiation
- Low energy
- Cannot produce ions
- Example: visible light

Ionizing radiation
- High energy
- Can produce ions
- Examples: X-rays, nuclear radiation

2 X-rays

X-ray tube



Applications
X-ray imaging for
- medicine
- security
- industry
- archaeology



3 Nuclear radiation

Emitted spontaneously by the unstable nuclei of radioactive substances

α radiation
- Stream of helium nuclei
- Strong ionizing power
- Low penetrating power

β radiation
- Stream of fast-moving electrons
- Weak ionizing power
- High penetrating power

γ radiation
- High-frequency EM waves
- Very weak ionizing power
- Very high penetrating power

refer to p.24 for a more detailed summary

4 Nuclear radiation detectors



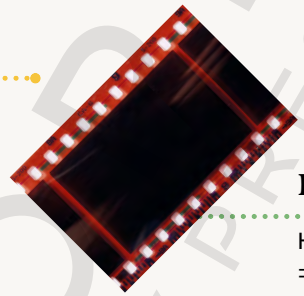
Spark counter

More ions formed
⟹ more sparks produced



Diffusion cloud chamber

Ionizing radiation produces white track along its path



Photographic film

Higher level of exposure
⟹ darker after developed



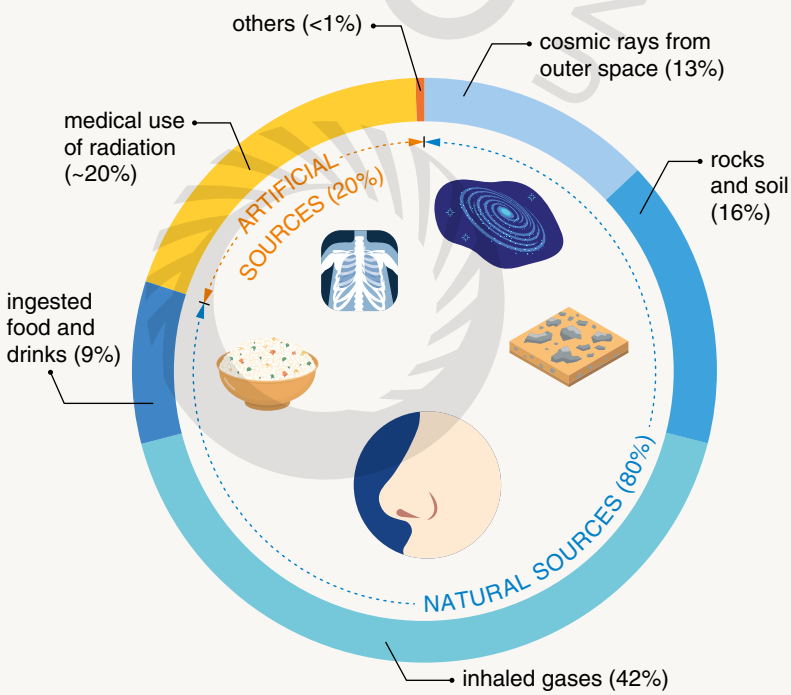
GM counter

More ions formed
⟹ higher count rate

Corrected count rate = measured count rate – background count rate

5 Radiation safety

Sources of radiation dose of an average person



Protection against radiation

- Minimize exposure time
- Stay away from radioactive sources
- Use barriers to shield radiation



- Keep radioactive sources out of environment

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Key terms

1 alpha radiation α 輻射 p.7
2 background radiation 本底輻射 p.9
3 beta radiation β 輻射 p.7
4 count rate 計數率 p.10
5 diffusion cloud chamber 擴散雲室 p.11
6 dose 劑量 p.30
7 equivalent dose 當量劑量 p.30
8 gamma radiation γ 輻射 p.7
9 Geiger-Müller counter 蓋革— 彌勒計數器 p.10
10 ion 離子 p.2
11 ionize 電離 p.2
12 ionizing power 致電離能力 p.13
13 ionizing radiation 致電離輻射 p.2
14 non-ionizing radiation 非電離輻射 p.2
15 nuclear radiation 核輻射 p.6
16 penetrating power 穿透能力 p.15
17 photographic film 感光底片 p.4
18 radioactive 放射性 p.6
19 radioactivity 放射現象 p.6
20 range 射程 p.14
21 right-angled fork track 直角分叉徑跡 p.20
22 sievert (Sv) 希沃特 p.30
23 spark counter 火花計數器 p.10

Revision 1

Multiple-choice questions

1 1.3 A doctor places a radioactive source in a person's body for medical check-up. The radiation emitted should have a weak ionizing power to minimize the damage caused to the person. Which of the following sources is the most suitable?

A α source B β source
C γ source D They are equally suitable.

***2** 1.2 Jacky sets up a diffusion cloud chamber. The observation is as shown in Figure a.



Fig a

At point *P*,

A ionizing radiation is emitted.
B ionizing radiation is stopped.
C ions are formed.
D ions are absorbed.

***3** 1.2 Which of the following statements about X-rays and γ radiation is **incorrect**?

A Both are ionizing radiation.
B Both are electromagnetic radiation.
C Both are spontaneously emitted from unstable nuclei.
D Both carry no charge.

***4** 1.3 In a laboratory, a GM counter is placed on a clear bench and it records a non-zero count rate. Which of the following is a possible source of the radiation detected by the counter?

A Radioactive source used for experiments
B Cosmic rays from outer space
C Walls of the laboratory
D All of the above

***5** 1.2 A right-angled fork track is formed when an α source is placed in a cloud chamber filled with helium gas. What conclusion can be drawn?

A An α particle is positively charged.
B An α particle has the same mass as a helium nucleus.
C An α particle consists of two protons and two neutrons.
D The ionizing power of α radiation is the strongest among nuclear radiations.

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★6 1.2 Daisy uses a GM counter to study a radioactive source. The background count rate is measured to be about 29 min^{-1}. When she places the source 10 cm from the GM tube, the count rate increases to 132 min^{-1}. Which of the following statements is correct?

A The source emits α radiation only.

B The source emits α radiation and β radiation.

C The source does not emit α radiation.

D She cannot tell if the source emits α radiation.

★7 1.2 A radioactive source is placed very close to a GM tube which is connected to a counter. Different absorbers are placed in turn between the source and the GM tube (Fig b).

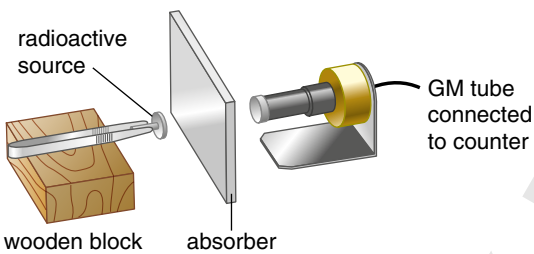


Fig b

The count rates measured for different absorbers are recorded in Table a.

Absorber	Count rate / min^{-1}		
—	1360	1382	1346
A sheet of paper	452	464	470
5-mm aluminium	445	467	473
25-mm lead	241	236	238

Table a

What type(s) of radiation does the source emit?

A α only B γ only

C α and γ only D β and γ only

★8 1.3 Which of the following actions can reduce the exposure to ionizing radiation?

(1) Exercise often

(2) Avoid unnecessary X-ray scan

(3) Maintain good ventilation in an indoor environment

A (2) only B (3) only

C (1) and (2) only D (2) and (3) only

★9 1.2 An α source is placed between two conducting plates (Fig c).

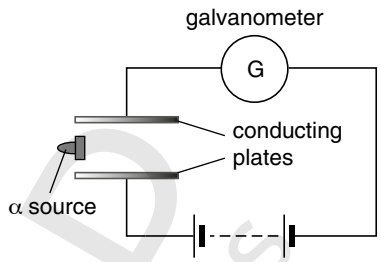


Fig c

The pointer of the galvanometer deflects. Which of the following statements is/are correct?

(1) A current is detected because α particles are attracted to one of the plates.

(2) No current will be detected if the set-up is placed in a vacuum.

(3) If the α source is replaced with a β source, the current flowing through the circuit is likely to be smaller.

A (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★10 1.2

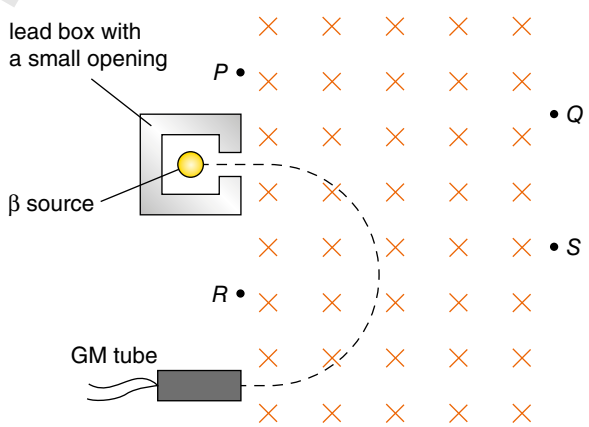


Fig d

A β source is placed in front of a uniform magnetic field pointing into page. Maximum reading is recorded by the GM tube at the position shown in Figure d. If the β source is replaced by an α source, at which position would the GM tube record a maximum reading?

A *P* B *Q*

C *R* D *S*

Q14 Exam Report: Just over one-third of the candidates had taken into consideration the possible range of radiation in the air.

***11** 1.2 Which of the following statements about detectors of nuclear radiation is/are correct?

(1) A GM counter can detect background radiation.

(2) Due to background radiation, tracks can be observed even if there is no radioactive source in a cloud chamber.

(3) A photographic film wrapped in thick paper can be used to detect any type of nuclear radiation.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

****12** 1.2 The following set-up is used to investigate the deflection of β radiation in a uniform magnetic field (Fig e). A GM tube connected to a counter measures the count rate as it moves along arc *PQ*.

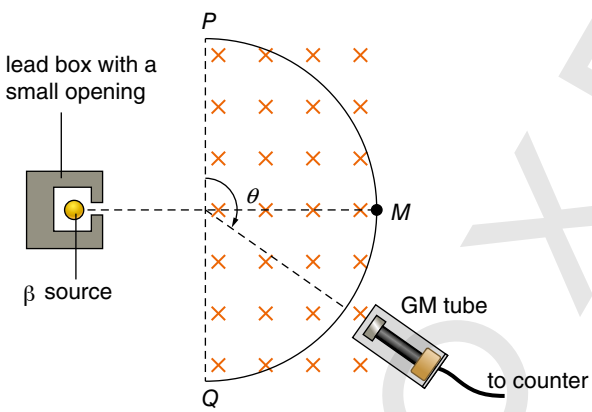
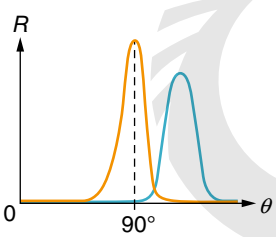


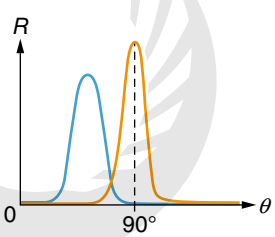
Fig e

Which of the following correctly shows the graphs of the corrected count rate *R* against θ when the magnetic field is in place (blue line) and not in place (orange line)?

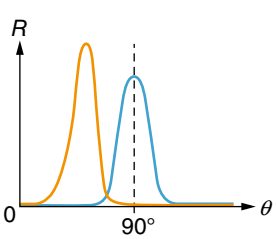
A



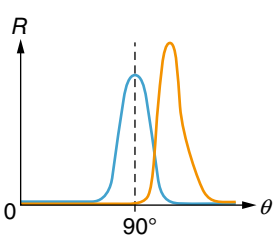
B



C



D



13 1.2 **HKDSE Practice Paper 2012 Paper 1A Q35**

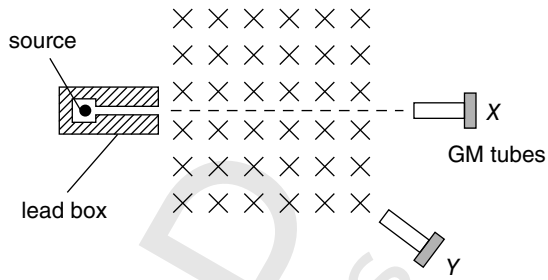


Fig f

A radioactive source is placed in front of a uniform magnetic field pointing into the paper as shown in Figure f. The count rates recorded by the GM tubes at *X* and *Y* are 101 counts per minute and 400 counts per minute respectively. Which of the following deductions must be correct?

A The source does not emit α radiation.

B The source emits β radiation.

C The source emits γ radiation.

D The background count rate is about 100 counts per minute.

14 1.2 **HKDSE 2015 Paper 1A Q32**

Workers in nuclear power plants wear clothes with film badges to measure the dosage of radiation received over a period of time. Which type of radiation below **cannot** be monitored by the film badges?

A α radiation (35%) B β radiation (2%)

C γ radiation (27%) D X-rays (36%)

15 1.2 **HKDSE 2020 Paper 1A Q30**

The background count rate in an experiment is determined using a GM counter. Four readings of the count rate in each minute are taken. Which set of readings below is the most probable?

	1st minute	2nd minute	3rd minute	4th minute
A	5	62	8	69
B	40	40	40	40
C	60	50	30	20
D	29	26	31	35

(63%)

Conventional questions

★16 1.2 A radioactive source is placed 3 cm in front of a GM tube, which is connected to a counter (Fig g). The count is recorded after 3 minutes and the counter is reset. These procedures are repeated with different absorbers placed between the source and the GM tube.

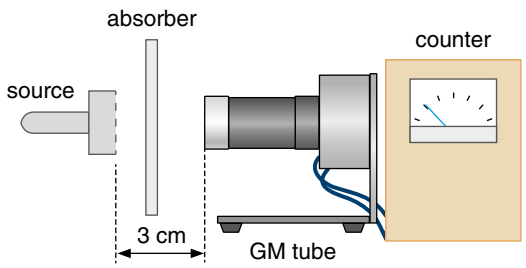


Fig g

The readings for each absorber are taken and recorded in Table b:

Absorber	Count
—	744
Paper	315
5-mm aluminium	75
25-mm lead	77
50-mm lead	73

Table b

(a) State one safety precaution that should be taken in the experiment. (1 mark)

(b) (i) Estimate the background count rate. (1 mark)

(ii) Estimate the corrected count rate when no absorber is used. (2 marks)

(c) Does the source emit β radiation? Explain briefly. (2 marks)

★17 1.2 In a physics lesson, your teacher gives you a pure β source and asks you to verify that β particles are negatively charged by using the apparatus shown in Figure h. Describe the procedures of the experiment with the aid of suitable diagrams and state the expected results. (7 marks)

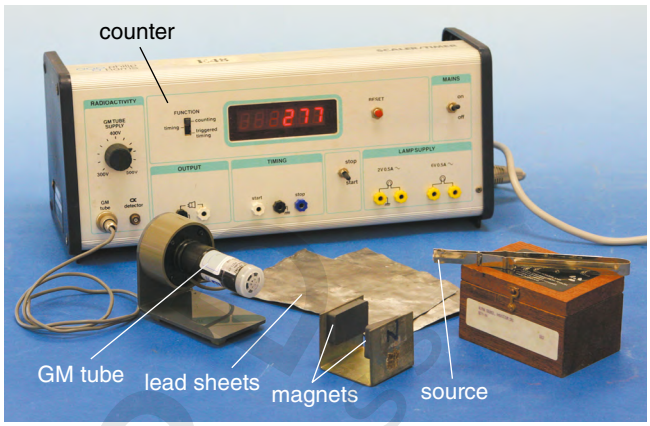


Fig h

★18 1.2 Radiation workers are required to wear film badges to monitor their exposure to radiation. A film wrapped in paper is embedded in the badge. Aluminium and lead sheets are also placed inside the badge (Fig i).

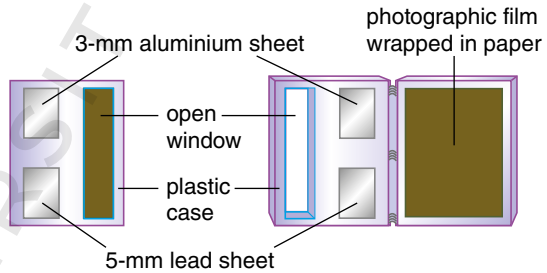


Fig i

The films worn by two workers, Ben and Collin, are developed. The degree of blackening on different regions of the films is shown in Table c.

Region on the film	Degree of blackening	
	Ben	Collin
Beneath open window	5	4
Beneath aluminium sheet	3	4
Beneath lead sheet	0	3

Note: 0 = not blackened, 5 = most blackened

Table c

(a) Why is the film wrapped? (1 mark)

(b) From the above data, explain whether Ben and Collin are definitely exposed to α, β or γ radiation. (6 marks)

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*19 1.2 A radioactive source emitting α and β radiations is placed in front of the two conducting plates that are connected to a power supply. The radiations split into two streams *X* and *Y* as they travel between the plates (Fig j).

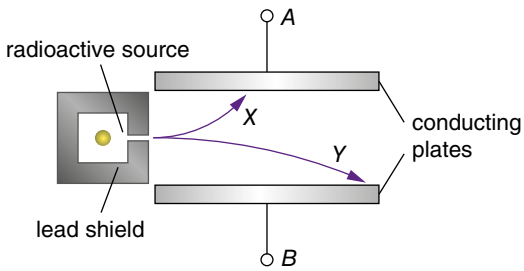


Fig j

(a) State the nature of α and β radiation. (2 marks)

(b) (i) What types of nuclear radiation are *X* and *Y*? Explain briefly. (4 marks)

(ii) State the polarities of terminals *A* and *B*. (1 mark)

(iii) The electric field is then replaced with a magnetic field. The radiation splits in a similar way as shown in Figure j in the magnetic field. What is the direction of the magnetic field? (1 mark)

*20 1.2 A moon rock sample was tested for radioactivity and the following data was obtained (Table d).

Detector	Position of detector and results recorded
GM counter	1 cm from source in air: high count rate behind tissue paper: count rate significantly reduced
Cloud chamber	Short and thick tracks only
Photographic film	Darkened even when shielded by a 5-mm lead sheet

Table d

(a) Does the moon rock emit α radiation? Explain briefly. (2 marks)

(b) Does the moon rock emit β radiation? Explain briefly. (2 marks)

(c) Does the moon rock emit γ radiation? Explain briefly. (2 marks)

*21 1.1 Read the following passage about **painting authentication** and answer the questions that follow.

In the past, many artists liked to recycle canvases by painting over a rejected work. To unravel the work underneath, one can carry out an X-ray scanning to the painting.

During scanning, the pigments on the canvas will partially absorb the incident X-rays. The degree of absorption depends on the pigment composition, concentration, thickness, etc. The unabsorbed X-rays are detected and forms the X-ray image.

Here is an example: Figure k (the original painting) shows a rainforest in good weather, while Figure l (the X-ray image) shows totally different weather.



Fig k



Fig l

(a) Suggest a reason why part *A* in Figure l is brighter than other parts. (2 marks)

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(b) Table e shows the main compositions of some pigments. According to your answer in (a), suggest a pigment colour used in part *A*. Explain briefly. (2 marks)

Pigment	Composition
black	carbon
white	lead

Table e

****22** 1.2 A radioactive source *R* is put in front of two oppositely charged parallel metal plates (Fig m). A GM tube is placed 15 cm away from *R*. The count rates at *X*, *Y* and *Z* are recorded.

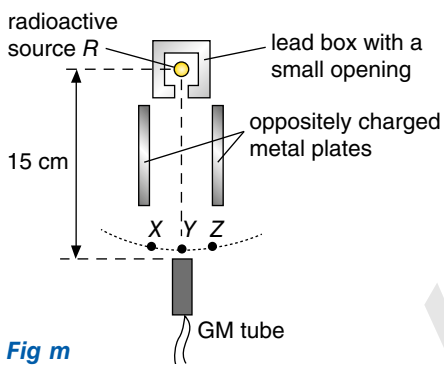


Fig m

Then the metal plates are removed, and the count rates are recorded again. The graphs below show the results (Figs n-p).

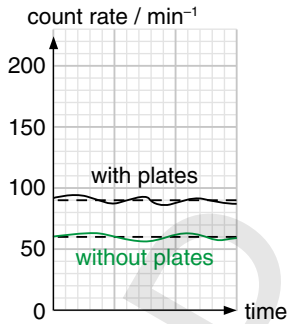


Fig n *Count rate at X.*



Fig o *Count rate at Y.*

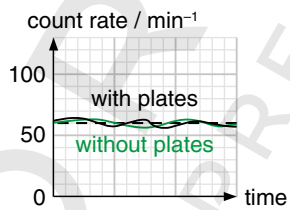


Fig p *Count rate at Z.*

(a) Estimate the background count rate. (1 mark)

(b) Can we determine whether *R* emits α radiation? Explain briefly. (2 marks)

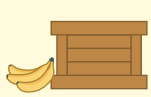
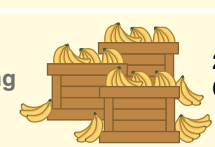
(c) Explain which kind of nuclear radiation makes the count rate at *X* increase in the presence of the metal plates. (2 marks)

(d) Estimate the count rates due to (i) β radiation and (ii) γ radiation at *Y* when the plates are removed. (2 marks)

Know more

Banana equivalent dose (p.30)

Bananas contain a trace amount of radioactive potassium-40. We are receiving a tiny dose of radiation (0.1 μSv) for eating each average-sized banana. This value turns out to be 1 Banana Equivalent Dose (BED), an informal but down-to-earth unit to give the idea of 'dose' to the public.

Eating one banana		1 banana 0.1 μSv = 0.0001 mSv
Dental X-ray		40 bananas 0.004 mSv
Single flight from Hong Kong to Japan		200 bananas 0.02 mSv
Annual dose from natural background radiation		24 000 bananas 2.4 mSv

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Know more

Wilhelm Röntgen (1845–1923) (p.3)

During the 1880s, there was great research interest in a device known as Crookes tube (Fig a), the predecessor of X-ray tube. One day in 1895, German physicist Wilhelm Röntgen (Fig b) covered his Crookes tube with a black cardboard and turn on the tube. He unexpectedly found that something invisible penetrated the cardboard and caused a fluorescent screen far away to glow. That was 'X-ray'.

In 1901, Röntgen was awarded the first Nobel Prize in Physics for discovering X-rays. In some languages, X-ray is even referred to as Röntgen rays.

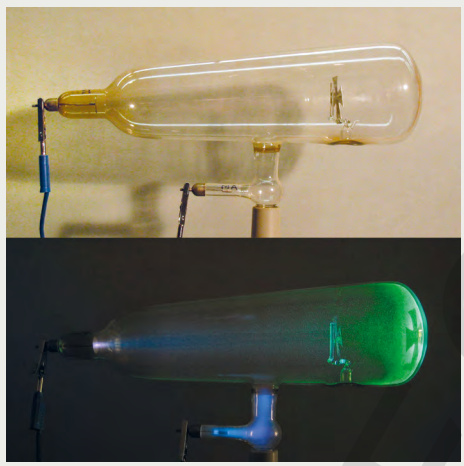


Fig a



Fig b

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Henri Becquerel (1852–1908) (p.6)

French physicist Henri Becquerel (Fig c) was an expertise on *phosphorescence*. After Röntgen had discovered X-rays, he decided to study whether there was any connection between phosphorescence and X-rays. Far beyond his expectation, he discovered the nuclear radiation.

Later, with the intensive research on radioactivity, Becquerel and the Curie couple (Fig d) discovered the origin of nuclear radiation — a spontaneous phenomenon happens at the nucleus. This laid the foundation of today's nuclear physics.

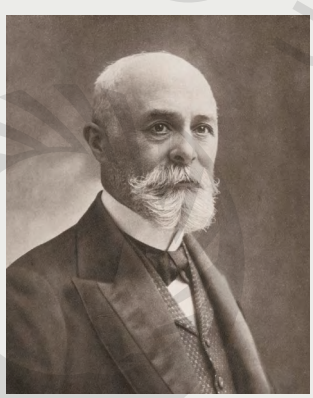


Fig c

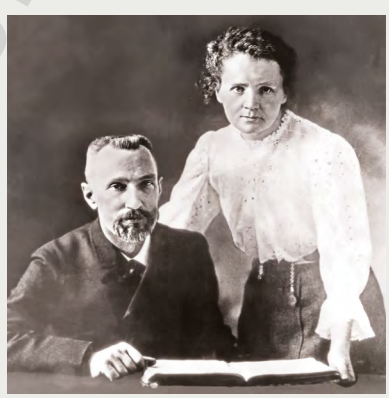


Fig d

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phosphorescence 磷光